

Security Lab Solution Document

Observability and Response Architecture Pt.2



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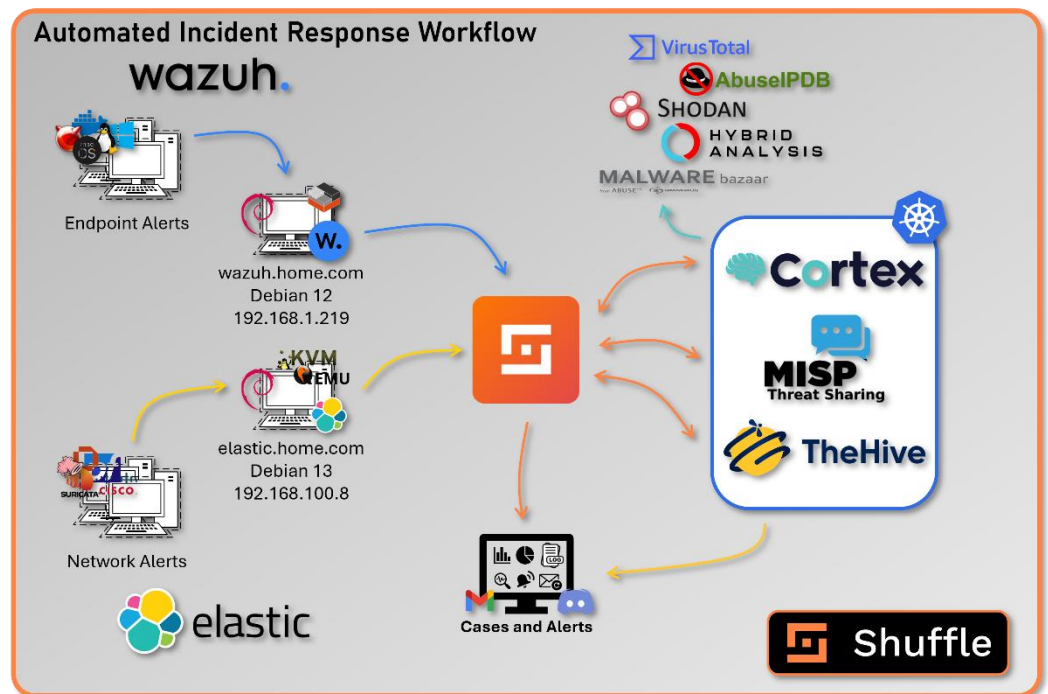
Observability and Response Architecture

Security Orchestration, Automation and Response (SOAR) Platform

Deployment Overview

The SOAR platform unifies case management, automated enrichment, threat intelligence sharing, and workflow automation across the entire security stack. It integrates SIEM alerts, EDR telemetry, threat intelligence feeds, and infrastructure controls into a coordinated response ecosystem. The platform consists of four primary applications: Shuffle, TheHive, Cortex, and MISP. Each fulfilling a specialized role within the incident response lifecycle. Together, they deliver

automated triage, enrichment, containment, and intelligence distribution, enabling rapid, repeatable, and scalable security operations.



Security Impact

The SOAR ecosystem dramatically reduces response time by automating repetitive tasks, enriching alerts with contextual intelligence, and orchestrating multi-tool actions. Automated workflows ensure consistent triage, reduce human error, and provide structured case management. Integrated threat intelligence enhances detection accuracy, while automated containment actions limit adversary dwell time. The platform provides full visibility into incident lifecycle metrics, enabling continuous improvement of SOC processes.

Deployment Rationale

Modern SOC operations require more than detection, they require coordinated, automated response. This SOAR architecture mirrors enterprise-grade security operations by combining case management, enrichment engines, threat intelligence platforms, and automation pipelines. It demonstrates proficiency with incident response frameworks, multi-tool orchestration, and automated decision-making. The platform also showcases the ability to integrate SIEM, EDR, IDS, firewalls, and external intelligence sources into a unified operational workflow.

Architecture Principles Alignment

Defense in Depth

- Multiple enrichment engines validate IOCs across independent sources
- Automated containment actions complement SIEM/EDR detections
- Case management, enrichment, and threat intelligence form overlapping response layers

Secure by Design

- Structured workflows enforce consistent triage and investigation procedures
- API-based integrations use authenticated, encrypted communication
- Automated playbooks reduce manual error and enforce policy compliance

Zero Trust

- No alert or IOC is implicitly trusted; all are enriched and validated
- Automated workflows enforce least-privilege actions and controlled response steps
- Continuous verification of threat intelligence before actioning

Shuffle – Security Automation Engine

Deployment Overview



Shuffle is the low-code/no-code automation engine that orchestrates workflows across all SOC tools, SIEM platforms, and security infrastructure. It serves as the automation backbone of the SOAR ecosystem, enabling rapid integration, event-driven workflows, and multi-tool response actions. Shuffle processes alerts from Wazuh, Splunk, Elastic, and other sources, automatically enriching them with intelligence from Cortex, VirusTotal, Shodan, and MISP before triggering case creation or containment actions.

Security Impact

- Automates triage, enrichment, and response actions to reduce MTTR
- Executes containment workflows (firewall blocks, host isolation, account disablement)
- Ensures consistent, repeatable response actions across all alerts
- Provides real-time notifications and workflow execution logs

Deployment Rationale

Shuffle demonstrates the ability to build enterprise-grade automation pipelines without requiring custom code. It integrates seamlessly with SIEM, EDR, threat intelligence, and case management systems, enabling end-to-end automated response. This deployment mirrors modern SOC automation strategies used to scale operations and reduce analyst workload.

Architecture Principles Alignment

Defense in Depth: Automates multi-layer response actions across EDR, firewalls, SIEM, and threat intelligence

Secure by Design: API-key isolation, encrypted communication, and controlled workflow execution

Zero Trust: Every alert is enriched and validated before action; no implicit trust in raw telemetry

Deployment Architecture:

Shuffle is deployed as a multi-container microservices architecture with the following components:

Component	Image	IP Address	Purpose
shuffle-frontend	shuffle-frontend:latest	192.168.200.41 (LB)	React-based web UI for workflow design, execution monitoring, and administration. Provides drag-and-drop workflow builder with 300+ pre-built app integrations.
shuffle-backend	shuffle-backend:latest	10.43.xxx.xxx (ClusterIP)	Go-based backend API handling workflow execution orchestration, app management, webhook processing, and user authentication. Coordinates communication between frontend and execution workers.

shuffle-opensearch	opensearch:3.2.0	10.43.xxx.xxx (ClusterIP)	OpenSearch database storing workflow definitions, execution history, app configurations, and audit logs. Provides full-text search capabilities for workflow debugging and audit trails.
shuffle-orborus	shuffle-orborus:latest	None (internal)	Worker orchestration daemon responsible for spinning up Docker containers to execute workflow app actions. Manages containerized execution environment ensuring isolation and scalability. Runs on Kubernetes nodes as DaemonSet.

External Access:

- Web Interface: <https://192.168.200.41> (ports 80/443)
- Webhook Receiver: <https://192.168.200.41/api/v1/hooks/> (for external trigger integrations)
- API Endpoint: <https://192.168.200.41/api/v1/> (RESTful API for programmatic workflow management)

Integration Points:

Shuffle serves as the central orchestration hub connecting:

- **TheHive:** Automated case creation, task assignment, observable enrichment, and case closure workflows
- **Cortex:** Trigger analysis jobs, retrieve results, execute responders based on analysis outcomes
- **MISP:** Automatic IOC submission, threat intelligence queries, event publishing, and feed synchronization
- **Wazuh EDR:** Alert triage, automated response actions (file quarantine, process termination), agent management
- **Elastic (ELK Stack):** Network alert triage via API webhook. pfSense, OPNsense and Suricata logs.
- **pfSense Firewall:** Automatic blocklist updates, firewall rule creation, VPN configuration changes
- **Suricata/Snort IDS:** Rule updates, signature deployment, detection tuning based on threat intelligence
- **Discord:** Real-time notifications, alert distribution, case status updates, workflow execution reports
- **Email:** Phishing analysis workflows, report distribution, executive summaries, alert forwarding
- **VirusTotal/AbuseIPDB:** Automated reputation checks, malware analysis, IP/domain enrichment

TheHive - Incident Response and Case Management

Deployment Overview

TheHive provides centralized case management, investigation workflows, and collaborative incident response. It aggregates alerts from SIEM and EDR platforms into structured cases, enabling analysts

to track tasks, observables, timelines, and response actions. TheHive acts as the command center for incident response, coordinating investigations across Cortex, MISP, Wazuh, and Shuffle.



Security Impact

- Centralizes all incidents into structured, auditable cases
- Enables collaborative investigations with task assignments and timelines
- Correlates alerts from Splunk, Wazuh, and other sources into unified cases
- Provides metrics dashboards for MTBD, MTTR, and case volume trends
- Ensures consistent documentation for forensic and compliance requirements

Deployment Rationale

TheHive mirrors enterprise SIIRP (Security Incident Response Platform) capabilities, demonstrating proficiency with structured case management, workflow orchestration, and collaborative investigations. It provides the backbone for SOC processes, ensuring every incident is tracked, enriched, and resolved with full accountability.

Architecture Principles Alignment

- **Defense in Depth:** Combines SIEM, EDR, and threat intelligence alerts into multi-source cases
- **Secure by Design:** Role-based access, audit logs, and structured workflows
- **Zero Trust:** All observables validated through Cortex/MISP before actioning

Deployment Specifications:

- Container Image: `strangebee/thehive:5.5.13-1`
- Deployment Type: Helm chart (managed deployment)
- Replicas: 1 (single instance)
- External Access: LoadBalancer service at 192.168.200.33
 - Port 9000 (HTTP/HTTPS web interface)
 - Port 9095 (Kamon metrics for Prometheus integration)
- Persistent Storage: Backed by Cassandra and Elasticsearch for data durability
- Resource Allocation:
 - CPU: 2 cores
 - Memory: 4GB
 - Storage: 20GB (application data via Cassandra)

Key Capabilities:

- Case Management: Create, assign, and track security incidents with customizable workflows
- Task Orchestration: Break down investigations into actionable tasks with assignments and deadlines
- Observable Analysis: Submit IOCs (IPs, domains, hashes, URLs) to Cortex for automated analysis
- Alert Correlation: Aggregate alerts from SIEM (Splunk, Wazuh) into unified cases
- Collaboration: Team-based investigations with real-time updates and commenting
- Reporting: Generate executive summaries and technical reports from case data
- Metrics Dashboard: Track MTTD (Mean Time to Detect), MTTR (Mean Time to Respond), case volume trends

Integration Points:

- **Cortex:** Automated observable analysis and enrichment
- **MISP:** Bi-directional IOC sharing (export confirmed threats, import external intelligence)
- **Wazuh:** Automated case creation from EDR alerts (via Shuffle workflow)
- **Shuffle:** Workflow automation for alert triage and notification distribution
- **Discord:** Real-time notifications for high-severity cases
- **SMTP Relay:** Gmail notifications

Cortex - Observable Analysis and Active Response Engine

Deployment Overview



Cortex automates the analysis of observables (IOCs) using a wide range of analyzers and responders. It enriches IPs, domains, hashes, URLs, and files with intelligence from VirusTotal, AbuseIPDB, Shodan, OTX, and internal sources. Cortex responders execute automated containment actions such as firewall rule updates, blocklist modifications, and case escalations.

Security Impact

- Provides automated, multi-engine IOC analysis
- Executes rapid containment actions via responders
- Enhances detection accuracy through cross-source enrichment
- Supports dynamic malware analysis via sandbox integrations

Deployment Rationale

Cortex demonstrates advanced enrichment and automated response capabilities found in enterprise SOCs. Its analyzer/responder model enables modular, scalable intelligence processing and automated containment workflows.

Architecture Principles Alignment

- **Defense in Depth:** Multiple analyzer categories validate IOCs across independent sources
- **Secure by Design:** Controlled responder execution; strict API authentication
- **Zero Trust:** No IOC trusted without multi-engine validation

Deployment Specifications:

- Container Image: `thehiveproject/cortex:latest`
- Deployment Type: Kubernetes Deployment
- Replicas: 1
- External Access: LoadBalancer service at 192.168.200.40
 - Port 9001 (HTTP API and web interface)
- Backend Database: Elasticsearch (for job history and results)
- Resource Allocation:
 - CPU: 2 cores (scales with analyzer concurrency)
 - Memory: 4GB (increases with active jobs)
 - Storage: 10GB (job results and cache)

Analyzers Enabled:

- VirusTotal
- MISP
- AbuseIPDB
- Malware Bazaar
- Shodan
- Cyberchef
- Hybrid Analysis

Responders Enabled:

- VirusTotal Downloader
- Mailer
- Wazuh
- Shuffle
- N8n

MISP - Threat Intelligence Sharing Platform

Deployment Overview



MISP provides structured threat intelligence management, IOC sharing, and collaborative intelligence workflows. It stores, correlates, and distributes threat indicators across the SOAR ecosystem. MISP integrates with Cortex, TheHive, Shuffle, and external intelligence feeds to enrich alerts and cases with contextual threat data.

Security Impact

- Enhances detection accuracy through curated threat intelligence
- Enables bi-directional sharing of confirmed IOCs
- Correlates related events to identify campaigns and intrusion sets
- Supports automated enrichment and threat scoring

Deployment Rationale

MISP demonstrates proficiency with structured threat intelligence, IOC lifecycle management, and intelligence-driven detection. It mirrors enterprise CTI workflows where intelligence is continuously ingested, validated, enriched, and distributed across SOC tools.

Architecture Principles Alignment

- **Defense in Depth:** Adds intelligence-driven detection to complement SIEM/EDR telemetry
- **Secure by Design:** Signed events, role-based access, and controlled feed synchronization
- **Zero Trust:** All intelligence validated before distribution; no implicit trust in external feeds

Deployment Specifications:

MISP is deployed as a multi-container application stack with the following components:

Component	Image	IP Address	Purpose
misp-core	ghcr.io/misp/misp-docker/misp-core:latest	192.168.200.37 (LB)	Main MISP application (web UI, API, background workers)
misp-db	mariadb:10.11	10.43.151.59 (ClusterIP)	MySQL/MariaDB database for MISP data storage
misp-redis	valkey/valkey:7.2	10.43.131.214 (ClusterIP)	Redis cache for session management and job queuing
misp-modules	ghcr.io/misp/misp-docker/misp-modules:latest	10.43.234.246 (ClusterIP)	Expansion and enrichment modules for automated IOC enrichment
misp-guard	ghcr.io/misp/misp-docker/misp-guard:latest	10.43.97.195 (ClusterIP)	Security proxy protecting MISP core from malicious input

External Access:

- Web Interface: <https://192.168.200.37> (ports 80/443)

Supporting Infrastructure

Cassandra - Distributed NoSQL Database



Purpose: Scalable, fault-tolerant data store for TheHive case data, observables, and audit logs.

Deployment Specifications:

- Container Image: `cassandra:4.1.7`

- Deployment Type: StatefulSet (ensures stable network identity and persistent storage)
- Replicas: 1 (production would use 3+ node cluster for high availability)
- External Access: LoadBalancer service at 192.168.200.36
- Port 9042 (CQL native protocol)
- Persistent Volume: 100GB SSD-backed storage
- Replication Strategy: SimpleStrategy with replication_factor=1 (single node)

Configuration Highlights:

- Keyspace: TheHive data stored in dedicated keyspace with TTL policies for data retention
- Consistency Level: LOCAL_ONE (single node) / LOCAL_QUORUM (production multi-node)
- Backup Strategy: Daily snapshots via cron job, retained for 14 days
- Monitoring: Exposed JMX metrics for Prometheus scraping

Elasticsearch - Search and Analytics Engine



Purpose: Full-text search, log aggregation, and analytics for TheHive cases, Cortex job results, and observables.

Deployment Specifications:

- Container Image: `docker.elastic.co/elasticsearch/elasticsearch:9.2.2`
- Deployment Type: StatefulSet (single-node cluster)
- Replicas: 1
- External Access: LoadBalancer service at 192.168.200.34
- Port 9200 (HTTP REST API)
- Persistent Volume: 200GB SSD-backed storage
- Cluster Name: `soc-elasticsearch`
- Node Roles: master, data, ingest (combined role in single-node deployment)

Index Management:

- TheHive Indices: Cases, tasks, observables, audit logs with 90-day retention
- Cortex Indices: Analysis jobs, results, analyzer reports with 30-day retention
- Index Lifecycle Management (ILM): Automated rollover and deletion based on age/size
- Snapshots: Daily backups to NAS via snapshot repository

Performance Tuning:

- Heap Size: 4GB (50% of allocated memory)
- Shards: 1 primary shard per index (low-volume lab environment)
- Replicas: 0 (single node, no replication)
- Refresh Interval: 5 seconds (balance between real-time search and indexing performance)

OpenSearch - Shuffle Workflow Database



Purpose: Storage and retrieval of Shuffle workflow definitions, execution history, app configurations, and audit logs.

Deployment Specifications:

- Container Image: `opensearch:3.2.0`
- Deployment Type: StatefulSet (single-node cluster)
- Replicas: 1
- External Access: ClusterIP only (internal communication with Shuffle backend)
- Port 9200 (HTTP REST API)

- Persistent Volume: 100GB SSD-backed storage
- Cluster Name: shuffle-opensearch

Data Storage:

- Workflow Definitions: JSON representations of workflow configurations and app settings
- Execution History: Complete audit trail of all workflow runs with timestamps, inputs, outputs, and errors
- App Cache: Downloaded app containers and OpenAPI specifications for rapid workflow execution
- User Activity Logs: Authentication events, workflow modifications, permission changes

Performance Characteristics:

- Query Performance: Sub-second search for workflow debugging and historical analysis
- Retention: 90-day execution history (configurable based on storage capacity)
- Backup Strategy: Daily snapshots to NAS; integrated with Kubernetes backup automation

Integration Points Throughout the Document

Integration Points:

- TheHive: Bi-directional IOC exchange (TheHive cases → MISP events, MISP feeds → TheHive alerts)
- Cortex: MISP lookup analyzer queries threat intelligence database
- Wazuh: EDR threat intelligence module queries MISP for known malicious indicators

Use Cases and Workflows

Use Case/Workflow1: Automated Endpoint Threat Intelligence Pipeline (Shuffle)

Overview

SOAR workflow that automates Wazuh alert triage through TheHive case management, Cortex-based IOC enrichment, and multi-channel notifications.

Trigger: Wazuh webhook (Severity ≥ 7)

Tools: Wazuh EDR, TheHive, Cortex, VirusTotal, AbuseIPDB, Discord, Gmail

Workflow Execution

1. Alert Ingestion

- **Wazuh Alert Med** (Trigger): Receives security event via webhook
- **LogCollection** (Parser): Extracts structured alert data

2. TheHive Integration

- **CreateAlert:** Generates alert in TheHive with Wazuh event metadata
- **Add IP to Alert:** Injects IP observables from alert
- **Add Hash to Alert:** Injects file hash observables from alert
- **Create Case from Alert:** Promotes alert to investigable case

3. IOC Enrichment

- **Get Case Observable:** Retrieves all observables for analysis
- **Run Analyzer VirusTotal:** Executes VirusTotal_GetReport_3_1 via Cortex
- **Run Analyzer AbuseIPDB:** Executes AbuseIPDB_1_1 via Cortex
- **Get Cortex Results:** Retrieves analysis reports (JSON)
- **Add Observable to Case:** Attaches enrichment results to case

4. Notification

- **Send Discord Alert:** Pushes case summary to #soc-alerts channel

- **Gmail Relay Notification:** Emails SOC distribution list with case details

Use Case Example

Scenario: Wazuh detects suspicious process execution with SHA256 hash and external IP connection.

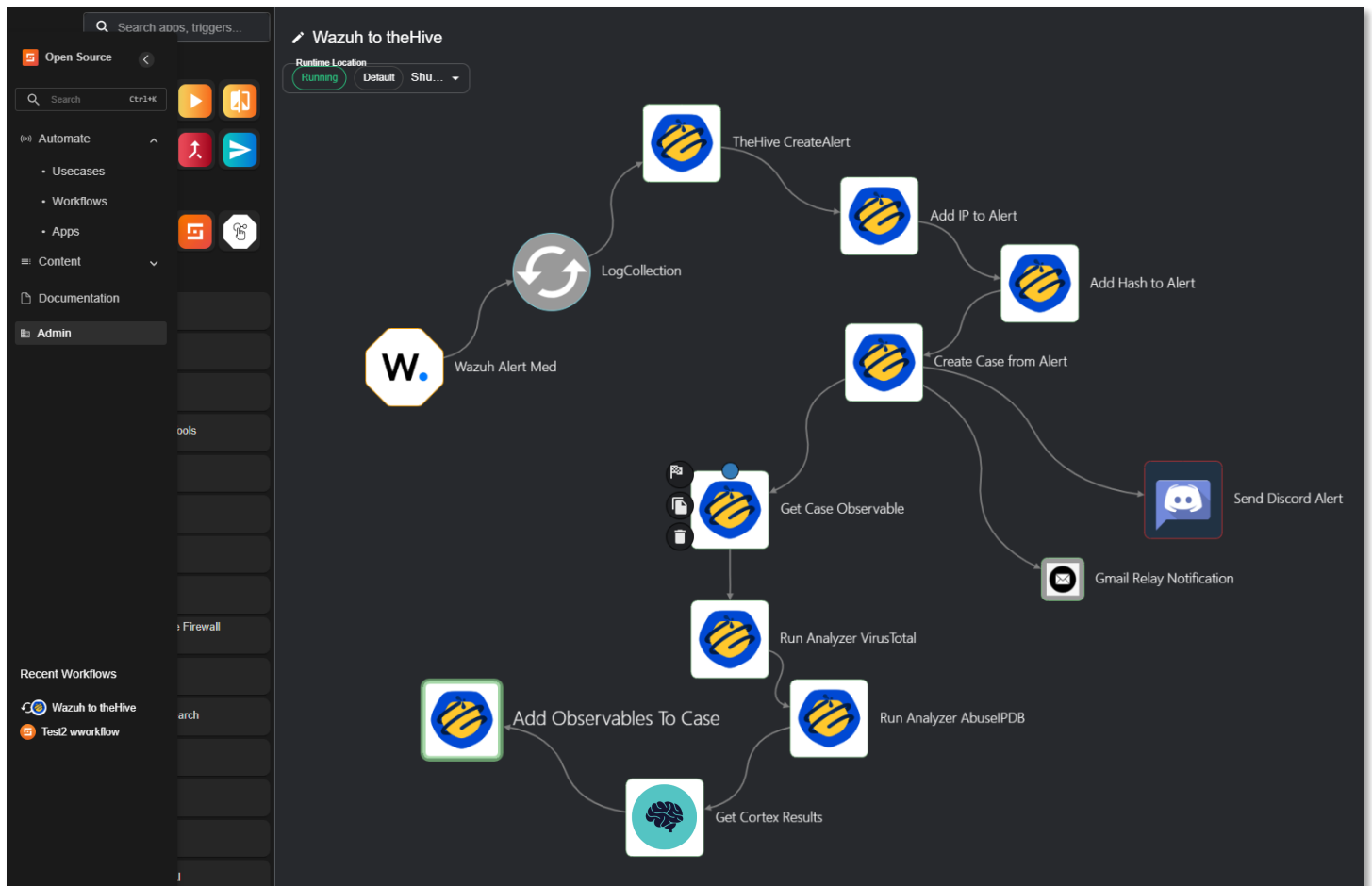
Automated Response:

1. Case created in TheHive: [Hostname] Suspicious PowerShell Execution
2. Cortex analysis results:
 - VirusTotal: 45/70 detections (Trojan.Downloader)
 - AbuseIPDB: IP flagged with 87% confidence score
3. Discord/Email alerts sent with enrichment summary
4. Analyst reviews case with pre-loaded context

Average Execution Time: 45-60 seconds

Workflow Screenshots:

Shuffle Workflow Overview



Wazuh Alert Events

	Feb 2, 2026 @ 10:50:52.488	DC01	File deleted.	7	553
	Feb 2, 2026 @ 10:50:52.485	DC01	File deleted.	7	553
☐ New New a few seconds ago M Wazuh Alert: DC01: File deleted. None None WAZUH_ALERT: DC01 WAZUH incident-2026-02-02T...					
	Feb 2, 2026 @ 11:06:22.500	unbound	sshd: Attempt to login using a non-existent user	5	5710
	Feb 2, 2026 @ 11:06:21.874	unbound	sshd: Attempt to login using a non-existent user	5	5710
	Feb 2, 2026 @ 11:06:20.944	unbound	PAM: User login failed.	5	5503
	Feb 2, 2026 @ 11:06:20.942	unbound	sshd: Attempt to login using a non-existent user	5	5710
	Feb 2, 2026 @ 11:06:20.942	unbound	sshd: Attempt to login using a non-existent user	5	5710

theHive Alerts

Alerts / WAZUH_ALERT (#incident-2...) / Description

Enter a case number

+ Create Case

id ~3420216

Created by Paul (Gmail)

Created at 03/02/2026 16:40

Import date 03/02/2026 16:41

SEVERITY:HIGH

TLP:GREEN

PAP:GREEN

Assignee

Paul (Gmail)

Source

WAZUH

Reference

incident-2026-02-03T16:40:44.042-0500

Type

WAZUH_ALERT

Occurred date

03/02/2026 16:40

Status

Imported

Case

#886

Import date

03/02/2026 16:41

Time metrics

Detection < 1 second

Triage 14 seconds

Acknowledge 14 seconds

Title

[OPNsense.internal] Multiple pfSense firewall blocks events from same source.

Tags

WAZUH OPNsense.internal mail=shad0w1t1a6@gmail.com

Description

###Wazuh Alert Details

Field	Value
Agent Name	OPNsense.internal
Host IP	192.168.1.201
Hostname	OPNsense.internal
User	****

Raw Log / Details

Feb 3 16:40:43 OPNsense.internal filterlog[37771]: 4,02f4bab031b57d1e30553ce08e0ec131,vtnet0,match,block,in,4,0x0,128,3115,0,none,17,udp,49,192.168.1.31,192.168.1.255,53846,32412,29

Summary

Not specified

Custom Fields

default (6)

Host_IP*	OS	OS_dist_version
192.168.1.201	FreeBSD	14
Add	Add	Add
PVE_id*	Recipient	hostname*
115	shad0w1t1a6@gmail.com	opnsense
Add	Add	Add

Observables (1)

TTPs (0)

Attachments

Similar Cases

Similar Alerts

Responders

History

Flags

Data type Value/Filename

TLP:AMBER

PAP:AMBER

ip 192[.]168[.]1[.]31

None

VT:GetReport="200 resolution(s)"

VT:GetReport="0/93"

Cortex Analyzers

Cortex

+ New Analysis

Jobs History

Analyzers

Responders

Organizations

Jobs History (61)

Data Types (15)

Job Type (2)

Analyzers (18)

Observable

Search for observable data

Search

Clear

Pagination

Status	Job details	TLP	PAP
Success	[hash] ac26bb93992d19b5e24ca5b7efa133afcd9ed3f9193fd9c75880bdb32ac0864 Analyzer: VirusTotal_GetReport_3_1 Date: a minute ago User: Homelab/thehive	TLP-AMBER	PAP-AMBER
Success	[ip] 173[.]68[.]121[.]223 Analyzer: AbuseIPDB_1_1 Date: 2 hours ago User: Homelab/thehive	TLP-AMBER	PAP-AMBER
Success	[ip] 192[.]168[.]1[.]31 Analyzer: VirusTotal_GetReport_3_1 Date: 2 days ago User: Homelab/thehive	TLP-AMBER	PAP-AMBER

theHive Cases

Cases

Enter a case number

+ Create Case

default*

Quick Filters

Export list

assignee: Paul (Gmail)

stage: any(InProgress,closed)

Status	Severity	#Number	Title	Details	Assignee
In progress	M	#780	[elastic] Host-based anomaly detection event (rootcheck).	Tasks Observables TTPs Linked Alerts	0 0 0 1
In progress	H	#767	[pihole] SCA summary: CIS Ubuntu Linux 22.04 LTS Benchmark v2.0.0.: Score less than 50% (45)	Tasks Observables TTPs Linked Alerts	0 0 0 1
In progress	M	#665	[OPNsense.internal] Multiple pfSense firewall blocks events from same source.	Tasks Observables TTPs Linked Alerts	0 1 0 1
In progress	L	#646	[stepca] Integrity checksum changed.	Tasks Observables TTPs Linked Alerts	0 1 0 1
False positive	M	#616	[stepca] Integrity checksum changed.	Tasks Observables TTPs Linked Alerts	0 1 0 1
In progress	L	#593	[DC01] Computer account added/changed/deleted	Tasks Observables TTPs Linked Alerts	0 0 0 1
In progress	H	#591	[K3s-worker] systemd: Service exited due to a failure.	Tasks Observables TTPs Linked Alerts	0 0 0 1
In progress	M	#580	[UbuntuVM1] New dpkg (Debian Package) installed.	Tasks Observables TTPs Linked Alerts	0 0 0 1
In progress	M	#5	Wazuh Alert: dockerVM2: Listened ports status (netstat) changed (new port opened or closed).	Tasks Observables TTPs Linked Alerts	0 0 0 1

Discord and Email Notifications

[OPNsense.internal] Multiple pfSense firewall blocks events from same source.

Delete

Archive

Report

Move to

Reply

Zoom

Quick steps

Read / Unread

Categorize

Flag / Unflag

Print

[OPNsense.internal] Multiple pfSense firewall blocks events from same source.

shad0w1t1a6@gmail.com

To: You

Tue 2/3/2026 12:39 PM

🔔

Security Operations Center Alert

Case Name:

[OPNsense.internal] Multiple pfSense firewall blocks events from same source.

Case #:

807

Severity:

2

Summary

###Wazuh Alert Details | Field | Value | | :-- | :-- | **Agent Name** | "OPNsense.internal" | | **Host IP** | "192.168.1.201" | | **Hostname** | "OPNsense.internal" | | **User** | **** | ### 📄 Raw Log / Details > Feb 3 12:23:37 OPNsense.internal filterlog[37771]: 4,,,02f4bab031b57d1e30553ce08e0ec131_vtnet0_match_block,in,4,0x0,,1,44609,0,none,2,igmp,40,192.168.1.31,224.0.0.22,datalength=16

Metadata Details

• Timestamp:

• Assignee: pnleone17@gmail.com

• Created By: pnleone17@gmail.com

Actions Recommended

1. Review case details in [TheHive Portal](#).

2. Verify automated Cortex analyzer results.

3. Acknowledge this alert in the SOC dashboard.

SECURITY OPERATIONS CENTER

-- Sent via Shuffle SOAR *

🔴

New Case Created: [dockerVM2] SCA summary: Center for Internet Security Debian Linux 13 Benchmark: Score less than 50% (38)

Description:

###Wazuh Alert Details | Field | Value | | :-- | :-- | **Agent Name** | dockerVM2 | | **Host IP** | 192.168.1.166 | | **Hostname** | "" | | **User** | "" | ### 📄 Raw Log / Details > None

Case #

Severity

Assignee

766

2

pnleone17@gmail.com

Created By

pnleone17@gmail.com

Sent via Shuffle SOAR *

🔴

New Case Created: [OPNsense.internal] Host-based anomaly detection event (rootcheck).

Description:

###Wazuh Alert Details | Field | Value | | :-- | :-- | **Agent Name** | OPNsense.internal | | **Host IP** | 192.168.1.201 | | **Hostname** | "" | | **User** | "" | ### 📄 Raw Log / Details > Interface 'vtnet0' in promiscuous mode.

Case #

Severity

Assignee

765

2

pnleone17@gmail.com

Created By

pnleone17@gmail.com

Sent via Shuffle SOAR *

🔴

New Case Created: [pihole] SCA summary: CIS Ubuntu Linux 22.04 LTS Benchmark v2.0.0.: Score less than 50% (45)

Description:

###Wazuh Alert Details | Field | Value | | :-- | :-- | **Agent Name** | pihole | | **Host IP** | 192.168.1.250 | | **Hostname** | "" | | **User** | "" | ### 📄 Raw Log / Details > None

Case #

Severity

Assignee

767

2

pnleone17@gmail.com

Created By

pnleone17@gmail.com

Sent via Shuffle SOAR *

Use Case/Workflow2: Automated Network Threat Intelligence Pipeline (Shuffle)

TBD – in progress Shuffle, Network alerting from Firewalls (pfSense, OPNsense, Fortigate), IPS/IDS Suricata and Snort (via Elasic), theHive alerting and case creation, cortex analysis, Discord and email notifications.

elastic

Find apps, content, and more.

AI Assistant

Discover

suricata alerts

Check out context-aware Discover

Unsaved changes

Try ES|QL

Inspect

Alerts

Untitled

Data view

Network-Activity-View

Filter your data using KQL syntax

Last 15 minutes

Documents (14,363)

Field statistics

	@timestamp	source.ip	destination.ip	destination.port	network.transport	data_stream.dataset	event.type	tags
	Feb 8, 2026 @ 10:01:12.000	192.168.1.253	224.0.0.18	(null)	carp	pfsense.log	[connection denied]	OPNsense
	Feb 8, 2026 @ 10:01:12.000	192.168.1.35	255.255.255.255	65,506	udp	pfsense.log	[connection denied]	OPNsense
	Feb 8, 2026 @ 10:01:12.000	192.168.1.199	224.0.0.22	(null)	igmp	pfsense.log	[connection denied]	OPNsense
	Feb 8, 2026 @ 10:01:12.000	192.168.1.1	224.0.0.1	(null)	igmp	pfsense.log	[connection denied]	OPNsense
	Feb 8, 2026 @ 10:01:11.000	192.168.1.126	10.20.0.1	6,556	tcp	pfsense.log	[connection allowed]	OPNsense
	Feb 8, 2026 @ 10:01:11.000	192.168.1.126	10.20.0.1	6,556	tcp	pfsense.log	[connection allowed]	OPNsense
	Feb 8, 2026 @ 10:01:11.000	192.168.1.126	224.0.0.251	5,353	udp	pfsense.log	[connection denied]	OPNsense
	Feb 8, 2026 @ 10:01:11.000	192.168.1.202	224.0.0.251	5,353	udp	pfsense.log	[connection denied]	OPNsense
	Feb 8, 2026 @ 10:01:11.000	192.168.1.253	224.0.0.18	(null)	carp	pfsense.log	[connection denied]	OPNsense
	Feb 8, 2026 @ 10:01:10.147	192.168.200.2	224.0.0.18	(null)	carp	pfsense.log	[connection allowed]	pfSense
	Feb 8, 2026 @ 10:01:10.147	192.168.2.2	224.0.0.18	(null)	carp	pfsense.log	[connection allowed]	pfSense
	Feb 8, 2026 @ 10:01:10.147	192.168.1.247	192.168.200.8	9,200	tcp	pfsense.log	[connection allowed]	pfSense
	Feb 8, 2026 @ 10:01:10.147	192.168.1.247	192.168.200.8	9,200	tcp	pfsense.log	[connection allowed]	pfSense
	Feb 8, 2026 @ 10:01:10.147	192.168.1.152	192.168.200.8	9,200	tcp	pfsense.log	[connection allowed]	pfSense
	Feb 8, 2026 @ 10:01:10.147	192.168.1.152	192.168.200.8	9,200	tcp	pfsense.log	[connection allowed]	pfSense

elastic

Find apps, content, and more.

AI Assistant

Discover

suricata alerts

Check out context-aware Discover

Unsaved changes

Try ES|QL

Inspect

Alerts

+

+

+

+

+

Untitled

Data view

Network-Activity-View

Filter your data using KQL syntax

Last 15 minutes

Refresh

Documents (22,206)

Field statistics

Columns

Sort fields

Search

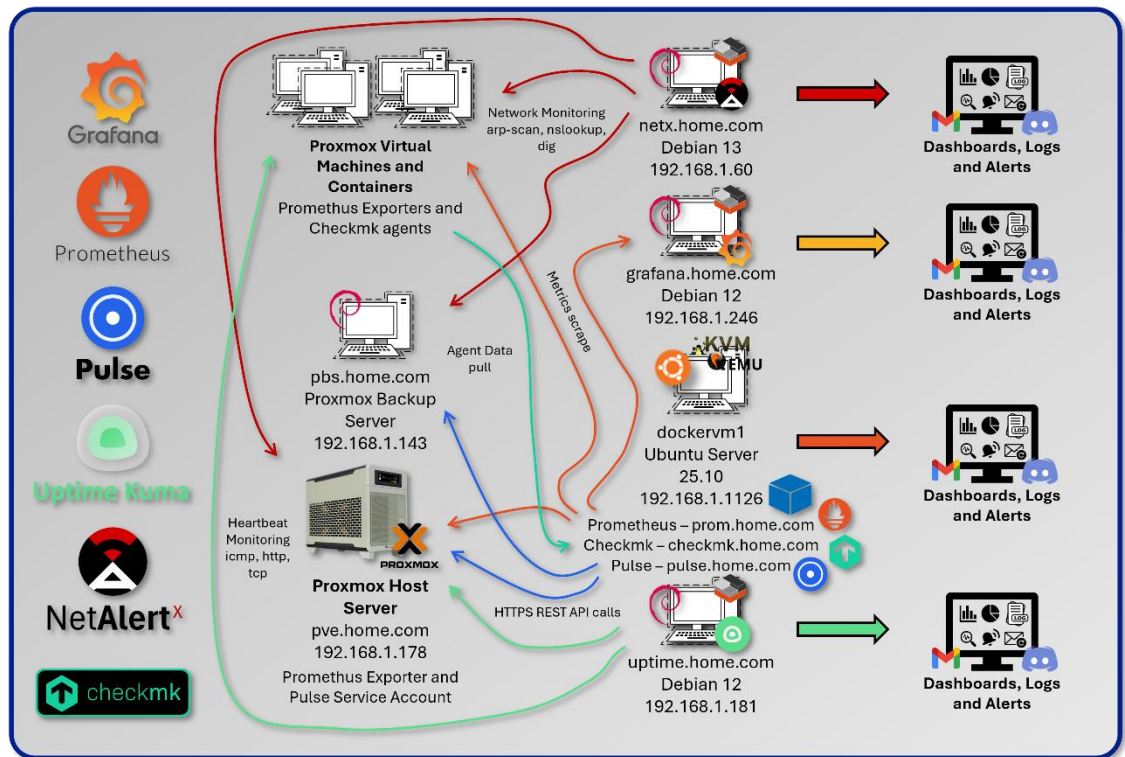
Menu

@timestamp	source.ip	destination.ip	destination.port	network.transport	data_stream.dataset	event.type	tags
Feb 8, 2026 @ 09:43:35.737	192.168.1.142	224.0.0.22	(null)	igmp	pfsense.log	[connection, denied]	[pfsense, forwarded]
Feb 8, 2026 @ 09:43:35.737	192.168.1.31	192.168.1.255	137	udp	pfsense.log	[connection, denied]	[pfsense, forwarded]
Feb 8, 2026 @ 09:43:35.000	192.168.1.142	224.0.0.22	(null)	igmp	pfsense.log	[connection, denied]	[opnsense, forwarded]
Feb 8, 2026 @ 09:43:35.000	192.168.1.31	192.168.1.255	137	udp	pfsense.log	[connection, denied]	[opnsense, forwarded]
Feb 8, 2026 @ 09:43:35.000	192.168.1.199	224.0.0.22	(null)	igmp	pfsense.log	[connection, denied]	[opnsense, forwarded]
Feb 8, 2026 @ 09:43:35.000	192.168.1.253	224.0.0.18	(null)	carp	pfsense.log	[connection, denied]	[opnsense, forwarded]
Feb 8, 2026 @ 09:43:35.000	192.168.1.31	192.168.1.255	137	udp	pfsense.log	[connection, denied]	[opnsense, forwarded]
Feb 8, 2026 @ 09:43:34.742	192.168.1.199	224.0.0.22	(null)	igmp	pfsense.log	[connection, denied]	[pfsense, forwarded]
Feb 8, 2026 @ 09:43:34.742	192.168.1.253	224.0.0.18	(null)	carp	pfsense.log	[connection, allowed]	[pfsense, forwarded]
Feb 8, 2026 @ 09:43:34.742	192.168.100.2	224.0.0.18	(null)	carp	pfsense.log	[connection, allowed]	[pfsense, forwarded]
Feb 8, 2026 @ 09:43:34.742	192.168.1.31	192.168.1.255	137	udp	pfsense.log	[connection, denied]	[pfsense, forwarded]
Feb 8, 2026 @ 09:43:34.742	192.168.200.2	224.0.0.18	(null)	carp	pfsense.log	[connection, allowed]	[pfsense, forwarded]
Feb 8, 2026 @ 09:43:34.742	192.168.2.2	224.0.0.18	(null)	carp	pfsense.log	[connection, allowed]	[pfsense, forwarded]
Feb 8, 2026 @ 09:43:34.742	192.168.1.33	192.168.200.39	3,001	tcp	pfsense.log	[connection, allowed]	[pfsense, forwarded]
Feb 8, 2026 @ 09:43:34.742	192.168.1.33	192.168.200.39	3,001	tcp	pfsense.log	[connection, allowed]	[pfsense, forwarded]
Feb 8, 2026 @ 09:43:34.742	192.168.1.31	224.0.0.251	5,353	udp	pfsense.log	[connection, denied]	[pfsense, forwarded]
Feb 8, 2026 @ 09:43:34.742	192.168.1.31	224.0.0.252	5,355	udp	pfsense.log	[connection, denied]	[pfsense, forwarded]

Observability Architecture

Deployment Overview

The monitoring stack provides comprehensive visibility into the health, performance, and availability of all systems across the lab environment. Unlike SIEM platforms that focus on security events, the monitoring layer captures operational telemetry, CPU, memory, disk, network throughput, service uptime, virtualization metrics, and application-level health checks. This multi-tool architecture spans time-series



metrics, heartbeat monitoring, deep infrastructure inspection, and hypervisor-specific analytics. Together, these systems form a unified observability fabric that enables proactive detection of performance degradation, service outages, and resource exhaustion before they escalate into security or operational incidents.

Security Impact

Monitoring is a critical component of operational security. Performance anomalies often precede or accompany security incidents, unexpected CPU spikes, abnormal network traffic, failing services, or resource exhaustion can indicate brute-force attempts, malware execution, or lateral movement. By correlating infrastructure telemetry with SIEM data, defenders gain a holistic view of system behavior. Monitoring also ensures high availability of security tooling itself (SIEM, EDR, firewalls, SOAR), preventing blind spots caused by service failures.

Deployment Rationale

Modern environments require more than log-based security analytics; they require continuous operational awareness. This monitoring architecture demonstrates proficiency with enterprise-grade observability tools, hybrid infrastructure monitoring, and multi-channel alerting. Each tool fulfills a distinct operational role—Prometheus for metrics, Grafana for visualization, Uptime Kuma for service uptime, Checkmk for deep infrastructure inspection, and Pulse for hypervisor-level awareness. The combined stack mirrors real-world NOC/SOC operations where performance, availability, and security telemetry converge.

Architecture Principles Alignment

Defense in Depth

- Multiple monitoring layers (metrics, uptime, hypervisor, infrastructure) ensure no single blind spot
- Redundant alerting channels prevent missed notifications
- Cross-tool correlation strengthens detection of operational anomalies

Secure by Design

- TLS-secured communication for metrics and alerting pipelines
- Segmented monitoring networks reduce exposure of sensitive telemetry

- Centralized dashboards enforce consistent visibility and reduce configuration drift

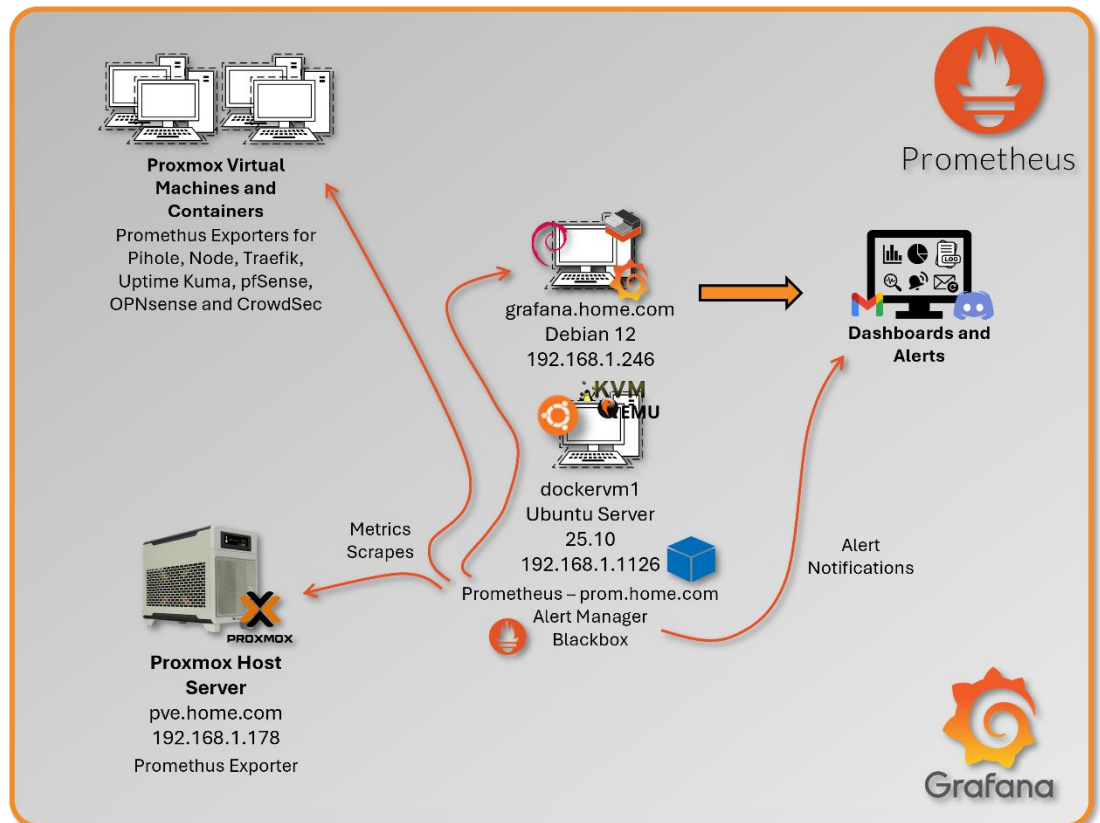
Zero Trust

- No host or service is implicitly trusted; all must report health and availability
- Continuous validation of service uptime and performance
- Alerting pipelines enforce accountability and traceability across all monitored systems

Prometheus & Grafana — Infrastructure Metrics Platform

Deployment Overview

Prometheus collects high-resolution time-series metrics from servers, containers, network devices, and applications. Grafana visualizes these metrics through customizable dashboards, providing real-time insights into system performance and long-term trends. This stack focuses on operational telemetry—CPU, memory, disk I/O, network throughput, container health, and service latency—complementing SIEM platforms by monitoring performance rather than security events.



Security Impact

- Detects resource exhaustion attacks (CPU spikes, memory leaks, disk saturation)
- Identifies anomalous network throughput that may indicate data exfiltration
- Monitors container and Kubernetes health for signs of compromise
- Provides early warning when security tools (SIEM, EDR, IDS) degrade or fail

Deployment Rationale

Prometheus and Grafana are industry-standard observability tools used across cloud-native and hybrid environments. Their inclusion demonstrates proficiency with metrics-driven monitoring, dashboard creation, alerting rules, and containerized instrumentation. This deployment mirrors enterprise monitoring strategies where Prometheus scrapes metrics and Grafana provides unified visualization.

Architecture Principles Alignment

- **Defense in Depth:** Metrics complement logs and endpoint telemetry for multi-layer detection
- **Secure by Design:** TLS-secured Prometheus endpoints; role-based Grafana access
- **Zero Trust:** Every host must expose metrics; no implicit trust in service health

Prometheus

Core Prometheus

- **Labels:** prom, pve, pi-hole
- **Intervals:** 1m

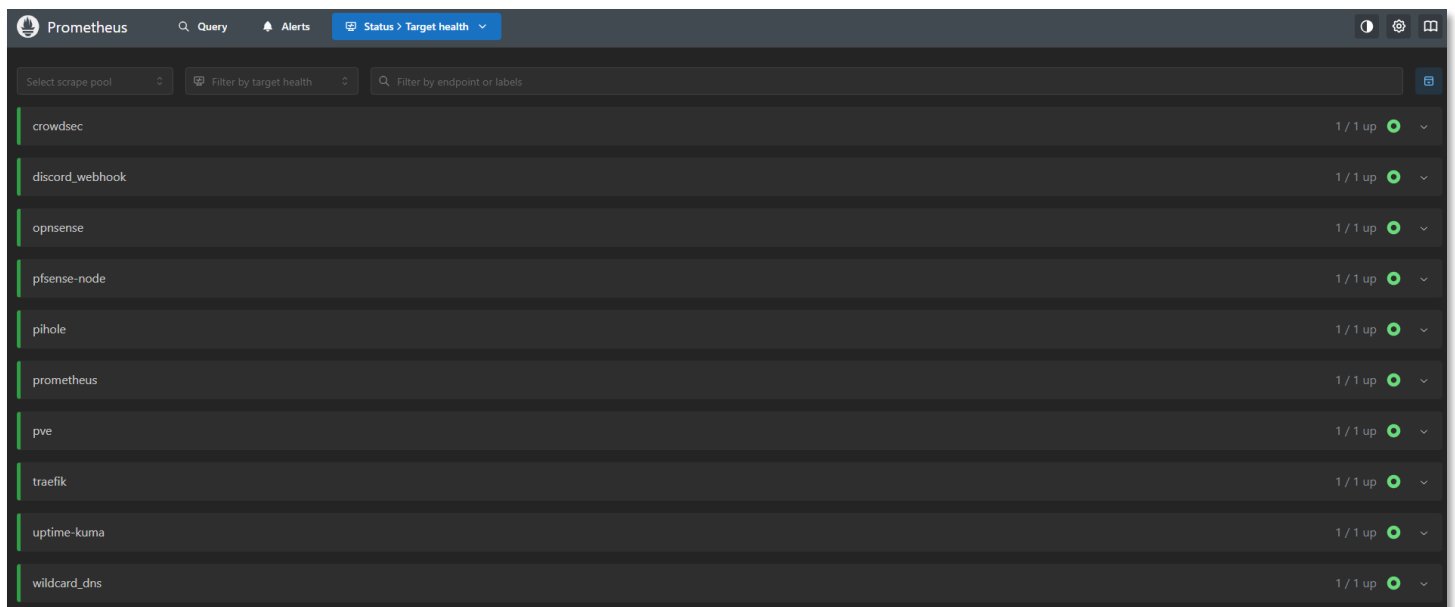
Alert Manager and Blackbox

- Notifications and Discord integration

Exporters

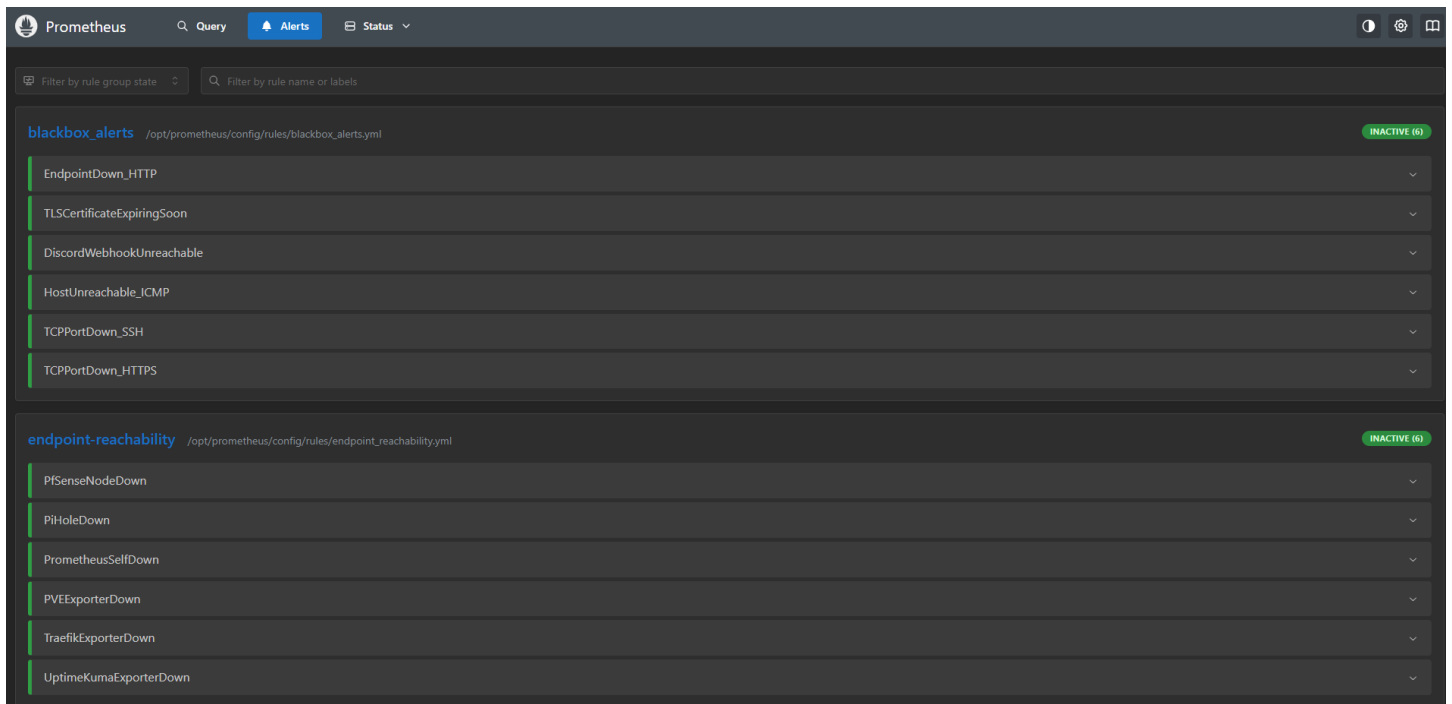
- **Node Exporter:** Linux hosts (PVE node, VMs, Docker hosts).
- **Proxmox Exporter:** prometheus-pve-exporter hitting the PVE API.
- **Traefik:** reverse proxy details.
- **pfSense Exporter:** system-level monitoring
- **OPNsense Exporter:** system-level monitoring
- **Uptime-Kuma:** Service uptime monitoring.
- **CrowdSec:** Engine metrics

Prometheus Target Health Status



Select scrape pool	Filter by target health	Filter by endpoint or labels
crowdsec		1 / 1 up ● ▾
discord_webhook		1 / 1 up ● ▾
opnsense		1 / 1 up ● ▾
pfsense-node		1 / 1 up ● ▾
pihole		1 / 1 up ● ▾
prometheus		1 / 1 up ● ▾
pve		1 / 1 up ● ▾
traefik		1 / 1 up ● ▾
uptime-kuma		1 / 1 up ● ▾
wildcard_dns		1 / 1 up ● ▾

Alert Manager and Blackbox Alerts



Grafana dashboards

Dashboards:

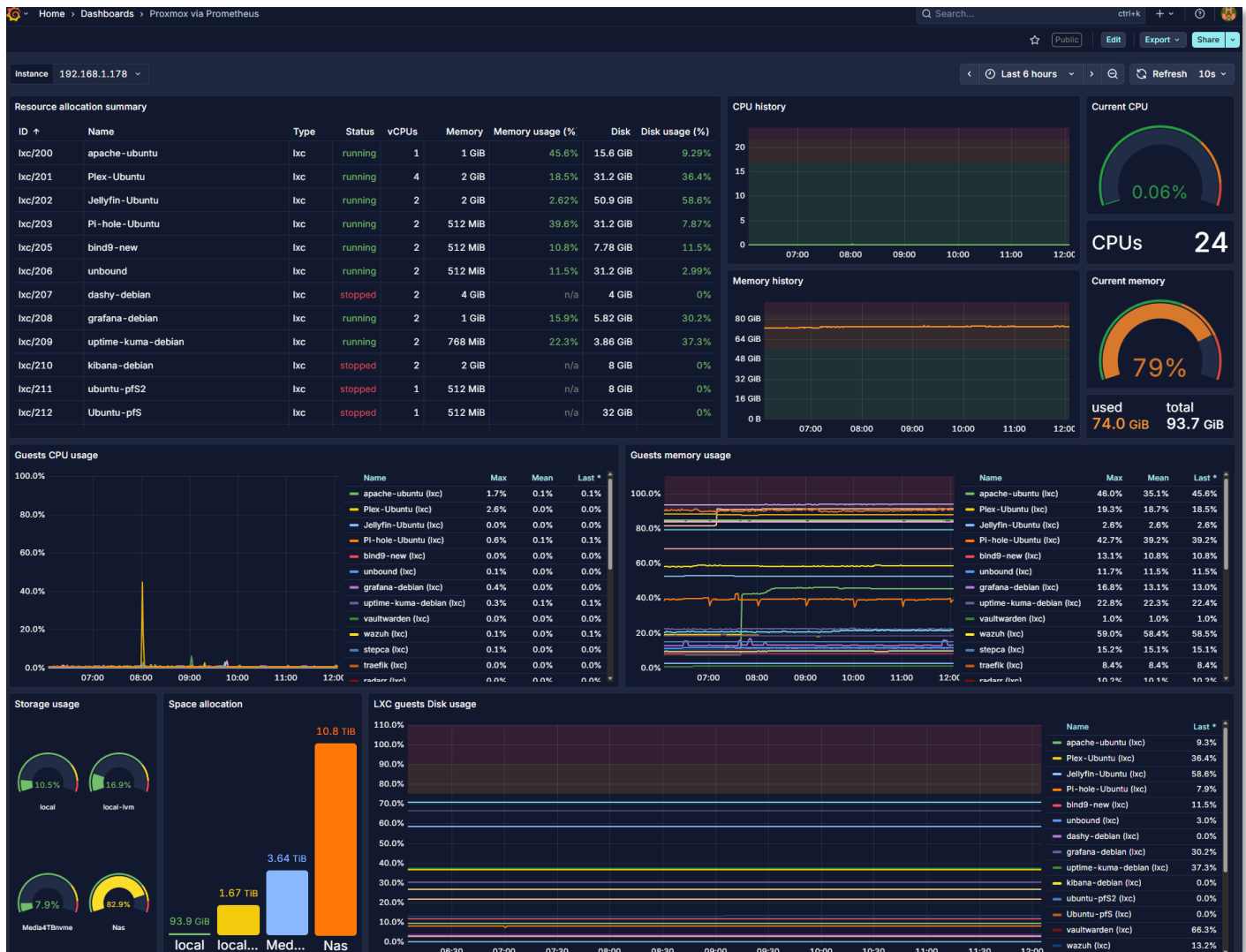
- **Node/Host Overview:** CPU, memory, FS, network, load (per host).
- **Proxmox:** Node status, VM CPU/mem/disk IO, cluster quorum.
- **Pi-hole:** Query volume, block rate, upstream latency, cache hit rate, top domains.
- **Traefik:** HTTP requests, slow services
- **pfSense:** system monitoring, CPU, RAM, traffic

Infrastructure Dashboards:

Dashboard Name	Data Source	Refresh Rate	Panels
Node Exporter Full	Prometheus	30s	CPU; RAM; Disk; Network; Load
Proxmox VE Overview	Prometheus	1m	Cluster status; VM metrics; storage
Traefik Overview	Prometheus	15s	Request rate; latency; error rate
pfSense System Metrics	Prometheus	30s	CPU; RAM; interfaces; throughput
Blackbox Exporter	Prometheus	1m	HTTP status; TLS cert expiry; latency

Proxmox VE

- **Exporter:** Run prometheus-pve-exporter as a service or container; use a least-privilege PVE API token.
- **Metrics:** Node/VM CPU, memory, storage pools, task failures, HA state.
- **Alerts:**
 - **Node pressure:** CPU > 85% for 10m, RAM > 90% for 5m.
 - **Storage:** Datastore free < 15%.
 - **VM health:** No metrics from a VM for > 5m (down).



Traefik

- **Exporter:** direct Prometheus integration.
- **Metrics:** instances, http requests per endpoint, application and http method, slow services.



pfSense

- **Exporter:** direct Prometheus integration.
- **Metrics:** CPU, RAM, Disk monitoring, network packets by interface, pkt/sec, load avg, traps and system calls.



Uptime Kuma - Service Availability Monitoring

Deployment Overview



Uptime Kuma provides heartbeat monitoring for all lab services using HTTP/S, TCP, and ICMP checks. Each monitored service is continuously probed for availability, latency, and response integrity. Any deviation triggers immediate alerts routed to dedicated Discord channels.

Security Impact

- Detects service outages caused by attacks, misconfigurations, or resource exhaustion
- Identifies unstable or degraded services before they impact security tooling
- Provides uptime baselines useful for correlating with SIEM events

Deployment Rationale

Service uptime is foundational to both operational stability and security visibility. Uptime Kuma offers a lightweight, flexible, and highly responsive monitoring layer that mirrors enterprise heartbeat monitoring systems.

Architecture Principles Alignment

- **Defense in Depth:** Adds availability monitoring to complement metrics and logs
- **Secure by Design:** Segmented monitoring probes reduce attack surface
- **Zero Trust:** No service is assumed healthy; continuous validation required



Edit Status Page

Go to Dashboard

All Systems Operational

Platform

100% Proxmox VE



91.25% Proxmox Backup Server



100% pfSense-A



100% OPNSense



99.86% Portainer



99.87% Traefik



100% Unbound



100% Bind



99.86% StepCA



99.86% Windows Server 2022



100% K3s - MetalLB



100% K3s Control Red Hat 10



100% K3s Worker Red Hat 10



100% cloudflare DDNS



100% Cloudflared - Traefik



100% Traefik-outpost



100% Whoami



100% Safeline WAF



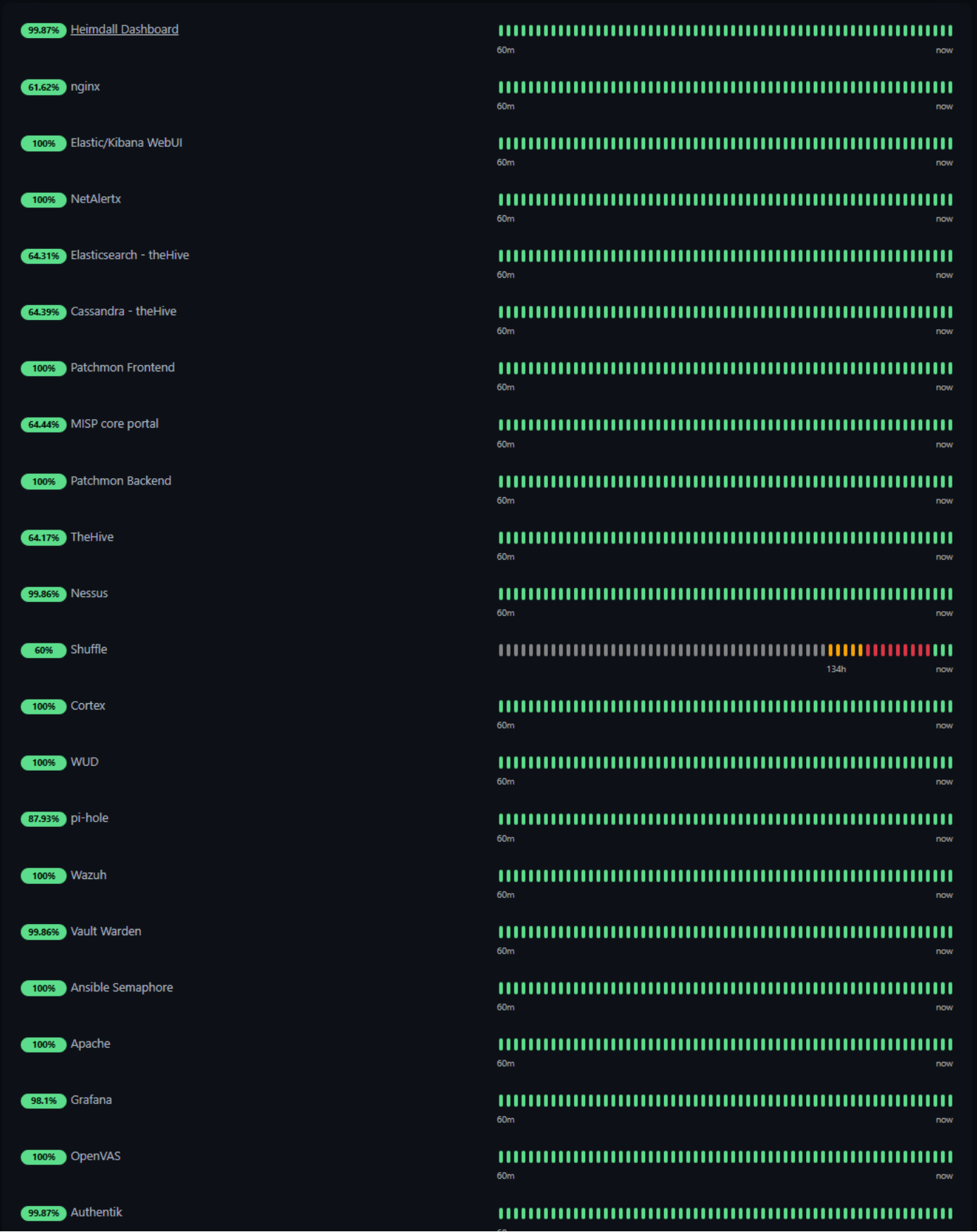
100% Smtip relay



100% Windows 2025 Server



Services



Checkmk – Deep Infrastructure Monitoring

Deployment Overview

Checkmk delivers comprehensive monitoring across servers, applications, network devices, storage systems, and containerized workloads.



Security Impact

- Detects abnormal system behavior (high load, failing disks, service crashes)
- Monitors critical infrastructure supporting security tools
- Provides granular visibility into multi-layer dependencies

Deployment Rationale

Checkmk represents enterprise-grade monitoring with agent-based and agentless capabilities. Its inclusion demonstrates proficiency with large-scale infrastructure monitoring, rule-based alerting, and hybrid environment observability.

Architecture Principles Alignment

- **Defense in Depth:** Adds deep host-level inspection to complement Prometheus metrics
- **Secure by Design:** Encrypted agent communication; strict role-based access
- **Zero Trust:** Every host must report detailed health metrics continuously

High-Level Configuration

The Checkmk monitoring server runs as a Docker container on UbuntuVM1, which serves as the central monitoring engine for the entire homelab. UbuntuVM1 sits in the LAB_LAN1 VLAN and has direct or routed access to all monitored networks.

The Checkmk container communicates with monitored hosts using a mix of native Checkmk agents, SNMP, API integrations, and special agents depending on the platform.

Monitored Host Types & Integration Methods

Proxmox Host and Linux VMs/LXCs (Ubuntu, Debian, CentOS, Fedora, RHEL, Kali, ParrotOS)

Integration: Checkmk Agent (TCP/6556)

Linux hosts across the lab run the Checkmk agent directly, providing deep OS-level visibility:

- CPU, memory, load, filesystem
- Systemd services
- Docker containers
- Kubernetes node metrics (for k3s control/worker nodes)
- Application-specific checks (Grafana, Prometheus exporters, Vaultwarden, Traefik, Unbound, etc.)

These hosts are monitored over routed VLANs with firewall rules allowing TCP/6556 from UbuntuVM1

Windows Hosts (Win11, Server 2022, Server 2025)

Integration: Checkmk Windows Agent

Windows endpoints and servers use the native Checkmk Windows agent, providing:

- CPU, memory, disk, services
- Event log monitoring
- Windows Update status
- Domain controller checks (for DC01, DC02)

pfSense Firewall

Integration: Checkmk Agent via xinetd (TCP/6556)

pfSense does not support the Checkmk agent natively, so the agent is deployed manually and served through xinetd. The service is configured to start at boot using pfSense's shellcmd mechanism.

Monitored metrics include:

- System health (CPU, RAM, swap)
- Interface throughput
- Gateway status
- Package health (Unbound, Suricata, Snort, etc.)
- Filesystem usage

Cisco vIOS Routers (r1, r2)

Integration: SNMP v2c

Cisco vIOS routers are monitored using SNMP, providing:

- Interface status and traffic
- Errors/discards
- CPU and memory
- Routing table size
- ARP table
- Device uptime

VMware ESXi Host

Integration: VMware Special Agent (HTTPS/443)

The ESXi host is monitored using the VMware API via the Checkmk special agent. A dedicated read only user is created on the ESXi host, and Checkmk retrieves:

- Host CPU and memory
- Datastore usage
- VM inventory and power state
- NIC and vSwitch metrics

Communication Summary

Host Type	Integration	Protocol	Notes
Linux servers	Checkmk Agent	TCP/6556	Full OS visibility
Windows servers	Checkmk Agent	TCP/6556	Event logs, services
pfSense	Checkmk Agent via xinetd	TCP/6556	Custom deployment
Cisco vIOS	SNMP v2c	UDP/161	Interface + routing metrics
ESXi	VMware Special Agent	HTTPS/443	API-based monitoring
Proxmox	Checkmk Agent	TCP/6556	Full OS visibility
Applications	Agent + HTTP checks	TCP/6556 + HTTP	Service-level monitoring

Main dashboard

Monitor > Overview > Main dashboard

Dashboard Add Dashboards Display Help

Host statistics		Service statistics		Host Problems (unhandled)		
20 Up 0 In downtime 0 Unreachable 0 Down 20 Total		290 OK 0 In downtime 0 On down host 29 Warning 3 Unknown 32 Critical 354 Total		State	Host	Icons
					Age	Summary

Service problems (unhandled)					Events of recent 4 hours		
State	Host	Service	Icons	Summary	Time	Host	Service
CRIT	OfficePC	Filesystem I/		Used: 96.41% - 8.77 TiB of 9.10 TiB (warn/crit at 80.00%/90.0 trend per 1 day 0 hours: +1.05%, Time left until disk full: 3 day	116 s	fedora	NTP Time
CRIT	Traefik	Memory		Total virtual memory: 14.23% - 146 MiB of 1.00 GiB, Page tab 8.00%/16.00% used) CRIT, 7 additional details available	116 s	fedora	NTP Time
CRIT	OfficePC	Log Application		396 CRIT, 172 WARN messages (Last worst: "Sep 29 08:59:3 cue_streamdeck_plugin.exe, version: 1.7.250.0, time stamp: 1 version: 6.7.1.0, time stamp: 0x6626b493 Exception code: 0xc process id: 0x9c44 Faulting application start time: 0x1dc3140 C:\Users\pneol\AppData\Local\Elgato\StreamDeck\Plugins\Faulting module path: C:\Users\pneol\AppData\Local\Elgato\StreamDeck\Plugins\2fc427d-f69e-426b-b502-8fc1c30bb4d9 Faulting package full Offset: 212 microseconds, Time since last sync: 31 minutes 5t synchronization (hosts time is running ahead) CRIT, Stratum: 91.189.91.157	236 s	fedora	NTP Time
CRIT	Splunk	Systemd Timesyncd Time		Offset: 0.0133 ms, Stratum: 2, Time since last sync: 17 minutes 9 seconds (warn/crit at 17 minutes 5 seconds/1 hour 0 minutes) WARN	236 s	fedora	NTP Time
CRIT	Debian-Docker2	Docker node info		Offset: 0.0133 ms, Stratum: 2, Time since last sync: 17 minutes 9 seconds (warn/crit at 17 minutes 5 seconds/1 hour 0 minutes) WARN	6 m	CrowdSec	Memory
CRIT	OfficePC	Log Security		Total virtual memory: 11.79% - 121 MiB of 1.00 GiB, Page tables: 51.14% - 262 MiB of 512 MiB RAM (warn/crit at 8.00%/16.00% used) CRIT, 7 additional details available	6 m	CrowdSec	Memory
CRIT	OfficePC	Log System		Total virtual memory: 11.79% - 121 MiB of 1.00 GiB, Page tables: 51.14% - 262 MiB of 512 MiB RAM (warn/crit at 8.00%/16.00% used) CRIT, 7 additional details available	6 m	CrowdSec	Memory
CRIT	OfficePC	Log Intel Graphics Software		Version: 2.4.0p12, OS: linux, TLS is not activated on monitored host (see details) WARN, Agent plug-ins: 0, Local checks: 0	6 m	CrowdSec	Check_MK Agent
CRIT	DC01-Server2022	Log Application		Version: 2.4.0p12, OS: linux, TLS is not activated on monitored host (see details) WARN, Agent plug-ins: 0, Local checks: 0	8 m	CentOS	Memory
CRIT	DC01-Server2022	System Time		Total virtual memory: 6.17% - 63.2 MiB of 1.00 GiB, Page tables: 51.18% - 262 MiB of 512 MiB RAM (warn/crit at 8.00%/16.00% used) CRIT, 7 additional details available	8 m	CentOS	Memory
CRIT	Webserver	Memory		Total virtual memory: 6.17% - 63.2 MiB of 1.00 GiB, Page tables: 51.18% - 262 MiB of 512 MiB RAM (warn/crit at 8.00%/16.00% used) CRIT, 7 additional details available	8 m	CentOS	Memory
CRIT	Webserver	Filesystem /sys/firmware/efi/efivars		Version: 2.4.0p12, OS: linux, TLS is not activated on monitored host (see details) WARN, Agent plug-ins: 0, Local checks: 0	8 m	CentOS	Check_MK Agent
CRIT	Pihole-primary	Memory		Version: 2.4.0p12, OS: linux, TLS is not activated on monitored host (see details) WARN, Agent plug-ins: 0, Local checks: 0	8 m	CentOS	Check_MK Agent
CRIT	Pihole-primary	Filesystem /sys/firmware/efi/efivars		Version: 2.4.0p12, OS: linux, TLS is not activated on monitored host (see details) WARN, Agent plug-ins: 0, Local checks: 0	8 m	CentOS	Check_MK Agent
CRIT	Bind9	Memory		Version: 2.4.0p12, OS: linux, TLS is not activated on monitored host (see details) WARN, Agent plug-ins: 0, Local checks: 0	10 m	Wazuh	Check_MK Agent
CRIT	Bind9	Filesystem /sys/firmware/efi/efivars		Version: 2.4.0p12, OS: linux, TLS is not activated on monitored host (see details) WARN, Agent plug-ins: 0, Local checks: 0	10 m	Wazuh	Check_MK Agent
CRIT	Unbound	Memory		Version: 2.4.0p12, OS: linux, TLS is not activated on monitored host (see details) WARN, Agent plug-ins: 0, Local checks: 0	13 m	Vaultwarden	Check_MK Agent
CRIT	Unbound	Filesystem /sys/firmware/efi/efivars		Version: 2.4.0p12, OS: linux, TLS is not activated on monitored host (see details) WARN, Agent plug-ins: 0, Local checks: 0	13 m	Vaultwarden	Check_MK Agent
CRIT	Stepca	Memory		Version: 2.4.0p12, OS: linux, TLS is not activated on monitored host (see details) WARN, Agent plug-ins: 0, Local checks: 0	13 m	Ansible	Systemd Service Summary
CRIT	DC01-Server2022	Log		Version: 2.4.0p12, OS: linux, TLS is not activated on monitored host (see details) WARN, Agent plug-ins: 0, Local checks: 0	13 m	Ansible	Systemd Service Summary
CRIT	DC01-Server2022	Log		Version: 2.4.0p12, OS: linux, TLS is not activated on monitored host (see details) WARN, Agent plug-ins: 0, Local checks: 0	13 m	Ansible	Memory
CRIT	Traefik	Inte		Version: 2.4.0p12, OS: linux, TLS is not activated on monitored host (see details) WARN, Agent plug-ins: 0, Local checks: 0	13 m	Ansible	Check_MK Agent
CRIT	Debian-Docker2	Inte		Version: 2.4.0p12, OS: linux, TLS is not activated on monitored host (see details) WARN, Agent plug-ins: 0, Local checks: 0	13 m	Ansible	Check_MK Agent
CRIT	Ubuntu-Docker1	Sys		Version: 2.4.0p12, OS: linux, TLS is not activated on monitored host (see details) WARN, Agent plug-ins: 0, Local checks: 0	13 m	Ansible	Check_MK Agent
CRIT	Uptime_Kuma	Mer		Version: 2.4.0p12, OS: linux, TLS is not activated on monitored host (see details) WARN, Agent plug-ins: 0, Local checks: 0	13 m	Ansible	Check_MK Agent
CRIT	Ansible	Mer		Version: 2.4.0p12, OS: linux, TLS is not activated on monitored host (see details) WARN, Agent plug-ins: 0, Local checks: 0	13 m	Ansible	Check_MK Agent
CRIT	Ansible	Sys		Version: 2.4.0p12, OS: linux, TLS is not activated on monitored host (see details) WARN, Agent plug-ins: 0, Local checks: 0	13 m	Ansible	Check_MK Agent
CRIT	CentOS	Mer		Version: 2.4.0p12, OS: linux, TLS is not activated on monitored host (see details) WARN, Agent plug-ins: 0, Local checks: 0	13 m	Ansible	Check_MK Agent
CRIT	CrowdSec	Mer		Version: 2.4.0p12, OS: linux, TLS is not activated on monitored host (see details) WARN, Agent plug-ins: 0, Local checks: 0	13 m	Ansible	Check_MK Agent
UNKN	Debian-Docker2	Doc		Version: 2.4.0p12, OS: linux, TLS is not activated on monitored host (see details) WARN, Agent plug-ins: 0, Local checks: 0	13 m	Ansible	Check_MK Agent

All hosts

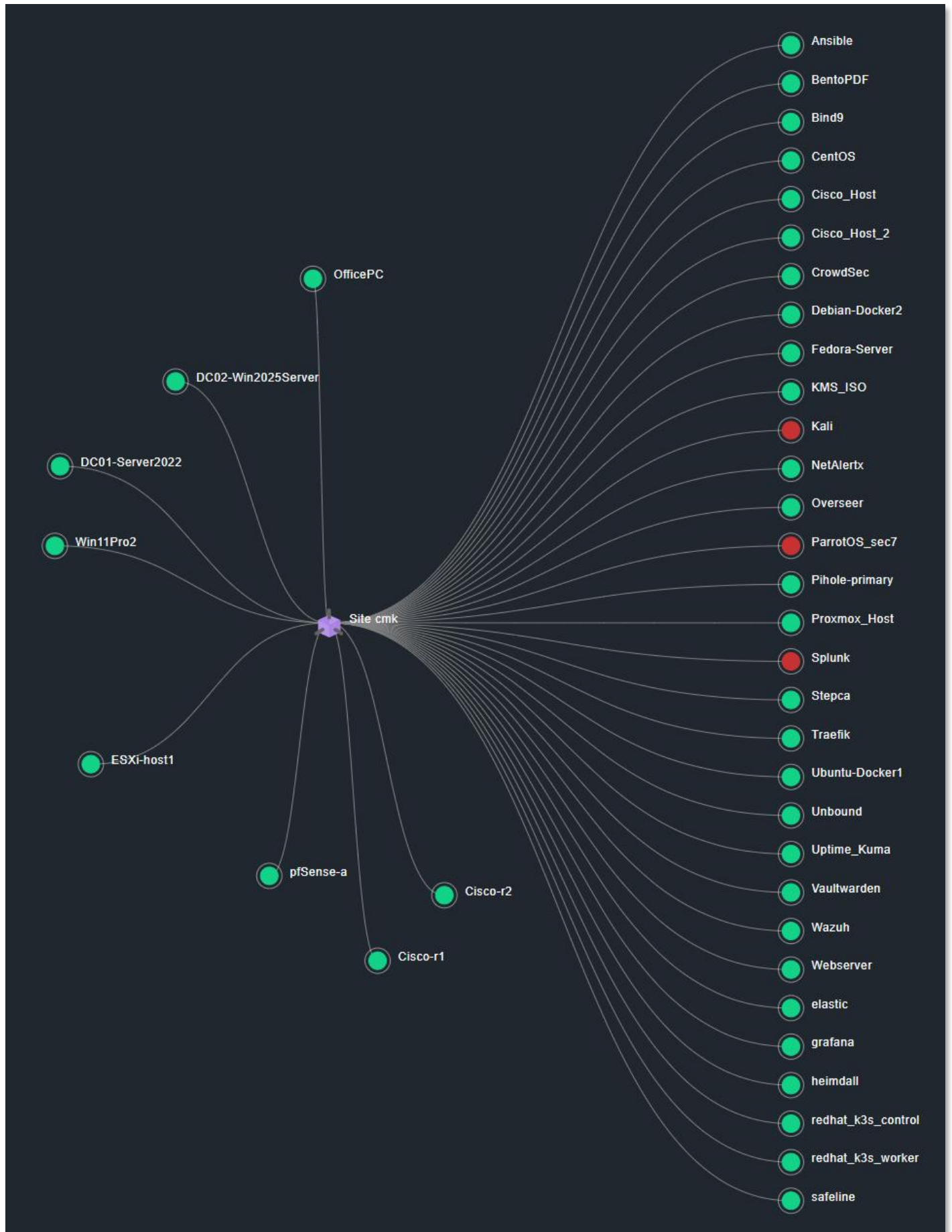
Monitor > Overview > All hosts

Commands Hosts Export Display Help

Acknowledge problems Schedule downtimes Filter Show checkboxes

State	Host	Icons	OK	Wa	Un	Cr
UP	Ansible		13	1	0	1
UP	BentoPDF		13	2	0	0
UP	Bind9		12	1	0	3
UP	CentOS		11	1	0	1
UP	Cisco-r1		10	1	0	0
UP	Cisco-r2		9	1	0	0
UP	Cisco_Host		13	1	0	2
UP	Cisco_Host_2		13	1	0	2
UP	CrowdSec		14	2	0	1
UP	DC01-Server2022		12	3	0	8
UP	DC02-Win2025Server		20	2	0	5
UP	Debian-Docker2		17	2	2	3
UP	elastic		2	1	0	0
UP	ESXi-host1		1	0	0	2
UP	Fedora-Server		17	2	0	0
UP	grafana		13	1	0	1
UP	heimdall		13	1	0	1
DOWN	Kali		15	1	0	1
UP	KMS_ISO		12	1	0	3
UP	NetAlertx		17	1	0	3
UP	OfficePC		17	3	1	8
UP	Overseer		16	1	0	2
DOWN	ParrotOS_sec7		11	1	0	2
UP	pSense-a		25	0	0	0
UP	Pihole-primary		13	3	0	3
UP	Proxmox_Host		2	1	0	0
UP	redhat_k3s_control		1	1	0	1
UP	redhat_k3s_worker		1	1	0	1
UP	safeline		23	1	0	0
DOWN	Splunk		17	1	0	2
UP	Stepca		13	1	0	1
UP	Traefik		16	0	0	3
UP	Ubuntu-Docker1		18	2	0	8
UP	Unbound		13	1	0	2
UP	Uptime_Kuma		12	1	0	2
UP	Vaultwarden		12	2	0	1
UP	Wazuh		13	2	0	0
UP	Webserver		9	4	0	3
UP	Win11Pro2		16	1	0	3

Topology Map



Pulse – Proxmox Virtual Environment and Backup Server Monitoring

Deployment Overview



Pulse provides real-time monitoring and alerting specifically for Proxmox VE and Proxmox Backup Server. It consolidates hypervisor metrics—VM performance, storage health, cluster state, backup integrity—into a unified dashboard, enabling rapid situational awareness and proactive incident response.

Security Impact

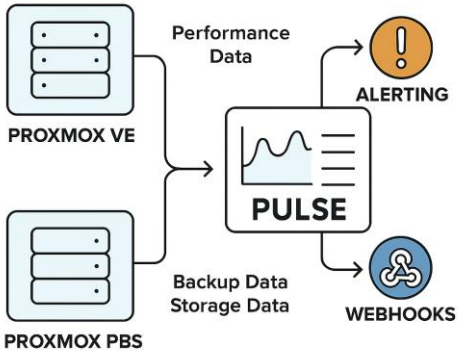
- Detects hypervisor-level anomalies that may indicate compromise or misconfiguration
- Monitors VM performance for signs of malicious activity
- Ensures backup integrity and cluster stability

Deployment Rationale

Virtualization is the backbone of the lab environment. Pulse provides hypervisor-specific insights that general monitoring tools cannot, mirroring enterprise virtualization monitoring platforms.

Architecture Principles Alignment

- **Defense in Depth:** Adds hypervisor-layer visibility beneath OS-level monitoring
- **Secure by Design:** Dedicated monitoring channel reduces exposure of Proxmox APIs
- **Zero Trust:** No VM or node is implicitly trusted; all must report health

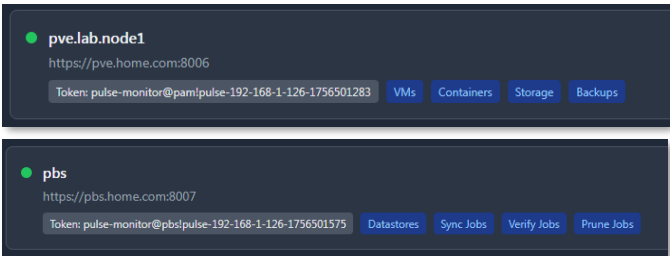


Lab Integration

In this lab environment, Pulse is connected to both the Proxmox VE cluster and the PBS node. Authentication is handled via a dedicated Proxmox service account configured with an API token. This account is assigned minimal, read-only permissions required to:

- Query node, VM, and container status.
- Retrieve backup job results.
- Report on storage utilization.

This least-privilege approach ensures that Pulse can monitor and report without having the ability to modify or disrupt the environment.



Built-In ↑	Name	Privileges
No	Dashboard	Datastore.Audit Ma
No	PulseMonitor	Sys.Audit
No	terraformprovision	Datastore.AllocateS VM.Config.Cloudini

Dashboards

The Pulse main dashboard provides a real-time operational view of the Proxmox environment, including:

- **Node statistics:** CPU load, memory consumption, and storage usage for each Proxmox VE host.
- **Guest statistics:** Resource usage for all deployed VMs and LXC.
- **Visual threshold indicators:** Color-coded gauges and bars highlight when usage approaches or exceeds configured limits.

- **Interactive webhooks:** Clicking on a monitored service or resource can open its corresponding Proxmox web portal page for direct management.

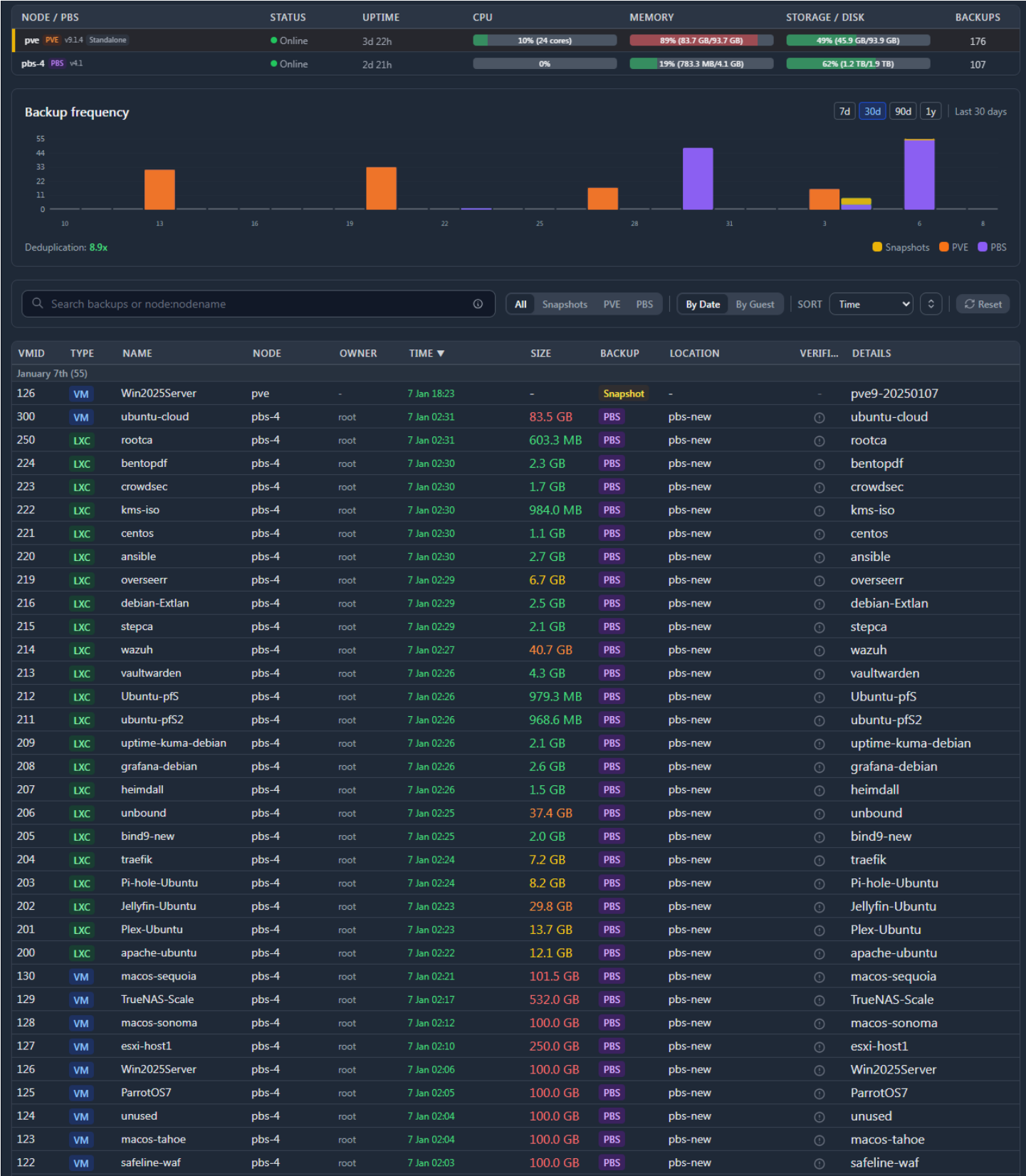
Example Use Cases in the Lab

- **Proactive capacity planning:** Identify nodes nearing CPU or memory saturation before performance degrades.
- **Backup compliance:** Immediate notification if a PBS backup job fails, allowing for same-day remediation.
- **Storage health:** Early warning when datastore usage trends toward capacity, preventing unexpected outages.

Node				Status	Uptime	CPU	Memory	Disk	VMS	Containers
pve PVE v9.1.4 Standalone				Online	3d 22h	11% (24 cores)	89% (83.3 GB/93.7 GB)	49% (45.9 GB/93.9 GB)	30	23
🔍 Search or filter... 🕒 All VMs LXCs All Running Stopped Grouped List 🔄 Reset										
Name	Type	VMID	Uptime ▼	CPU	Memory	Disk	Disk Read	Disk Write	Net In	Net Out
pve										
🟢 Pi-hole-Ubuntu 🟢🟢🟢 +4	LXC	203	3d 22h	0% (0.0/8 cores)	27% (439.3 MB/1.6 GB)	27% (8.4 GB/31.2 GB)	3 MB/s	28 KB/s	6 KB/s	2 KB/s
🟢 unbound 🟢🟢🟢 +1	LXC	206	3d 22h	2% (0.0/2 cores)	19% (97.3 MB/512.0 MB)	69% (43.5 GB/62.7 GB)	8 KB/s	19 MB/s	12 KB/s	65 KB/s
🟢 bind9-new 🟢🟢🟢	LXC	205	3d 22h	0% (0.0/2 cores)	15% (77.3 MB/512.0 MB)	26% (2.0 GB/7.8 GB)	0 B/s	797 B/s	5 KB/s	296 B/s
🟢 traefik 🟢🟢🟢 +1	LXC	204	3d 22h	1% (0.0/2 cores)	55% (558.6 MB/1.0 GB)	31% (7.3 GB/23.5 GB)	15 KB/s	997 B/s	30 KB/s	33 KB/s
🟢 pfsenseHA-A 🟢🟢🟢	VM	103	3d 22h	1% (0.0/2 cores)	51% (2.0 GB/4.0 GB)	-	0 B/s	136 KB/s	70 KB/s	69 KB/s
🟢 UbuntuVM1 🟢🟢🟢 +1	VM	100	3d 22h	5% (0.1/2 cores)	86% (10.1 GB/11.8 GB)	86% (47.1 GB/54.8 GB)	810 B/s	363 KB/s	50 KB/s	48 KB/s
🟢 Windows11Pro 🟢🟢🟢	VM	101	3d 22h	2% (0.1/4 cores)	38% (3.0 GB/8.0 GB)	30% (31.2 GB/105.3 GB)	0 B/s	20 KB/s	8 KB/s	61 B/s
🟢 win2022server 🟢🟢🟢 +1	VM	107	3d 22h	5% (0.1/2 cores)	89% (7.0 GB/7.9 GB)	-	5 KB/s	56 KB/s	12 KB/s	7 KB/s
🟢 Dockervm2 🟢🟢🟢 +1	VM	111	3d 22h	8% (0.2/2 cores)	95% (11.3 GB/11.9 GB)	68% (59.9 GB/88.1 GB)	0 B/s	128 KB/s	6 KB/s	17 KB/s
🟢 fedora-server 🟢🟢🟢	VM	113	3d 22h	0% (0.0/2 cores)	84% (1.7 GB/2.0 GB)	-	0 B/s	9 KB/s	81 B/s	20 B/s
🟢 opnsense 🟢🟢🟢	VM	115	3d 22h	2% (0.0/2 cores)	74% (1.5 GB/2.0 GB)	-	0 B/s	0 B/s	3 KB/s	4 KB/s
🟢 cisco-vios	VM	120	3d 22h	1% (0.0/2 cores)	6% (131.7 MB/2.0 GB)	-	0 B/s	0 B/s	2 KB/s	55 B/s
🟢 safeline-waf	VM	122	3d 22h	1% (0.0/4 cores)	81% (6.5 GB/8.0 GB)	14% (12.1 GB/88.1 GB)	0 B/s	51 KB/s	3 KB/s	1 KB/s
🟢 apache-ubuntu 🟢🟢🟢 +1	LXC	200	3d 22h	1% (0.0/1 cores)	44% (454.4 MB/1.0 GB)	78% (12.2 GB/15.6 GB)	0 B/s	797 B/s	7 KB/s	8 KB/s
🟢 grafana-debian 🟢🟢🟢	LXC	208	3d 22h	0% (0.0/2 cores)	10% (107.0 MB/1.0 GB)	46% (2.7 GB/5.8 GB)	997 B/s	7 KB/s	5 KB/s	3 KB/s
🟢 heimdall 🟢🟢	LXC	207	3d 22h	0% (0.0/2 cores)	8% (40.1 MB/512.0 MB)	5% (1.6 GB/31.2 GB)	0 B/s	0 B/s	73 B/s	6 B/s
🟢 stepca 🟢🟢🟢	LXC	215	3d 22h	0% (0.0/2 cores)	15% (75.5 MB/512.0 MB)	7% (2.1 GB/31.2 GB)	0 B/s	3 KB/s	89 B/s	47 B/s
🟢 vaultwarden 🟢🟢	LXC	213	3d 22h	0% (0.0/4 cores)	1% (49.7 MB/6.0 GB)	76% (4.4 GB/5.8 GB)	0 B/s	0 B/s	5 KB/s	13 B/s
🟢 uptime-kuma-d... 🟢🟢🟢 +1	LXC	209	3d 22h	1% (0.0/2 cores)	27% (208.6 MB/768.0 MB)	60% (2.3 GB/3.9 GB)	0 B/s	0 B/s	8 KB/s	3 KB/s
🟢 centos 🟢🟢🟢	LXC	221	3d 22h	0% (0.0/2 cores)	13% (66.5 MB/512.0 MB)	5% (1.2 GB/23.5 GB)	0 B/s	797 B/s	5 KB/s	33 B/s
🟢 ansible 🟢🟢🟢	LXC	220	3d 22h	0% (0.0/1 cores)	20% (102.5 MB/512.0 MB)	36% (2.8 GB/7.8 GB)	18 KB/s	4 KB/s	5 KB/s	74 B/s
🟢 overseerr 🟢🟢	LXC	219	3d 22h	0% (0.0/2 cores)	10% (211.6 MB/2.0 GB)	29% (14.0 GB/49.1 GB)	0 B/s	0 B/s	554 B/s	383 B/s
🟢 crowdsec 🟢🟢🟢	LXC	223	3d 22h	0% (0.0/2 cores)	23% (115.3 MB/512.0 MB)	24% (1.8 GB/7.8 GB)	0 B/s	3 KB/s	5 KB/s	184 B/s
🟢 esxi-host1	VM	127	3d 22h	2% (0.1/4 cores)	44% (3.2 GB/7.2 GB)	-	16 KB/s	12 KB/s	3 KB/s	20 B/s
🟢 pfsenseHA-B 🟢🟢🟢	VM	109	3d 20h	1% (0.0/2 cores)	40% (1.6 GB/4.0 GB)	-	0 B/s	213 KB/s	9 KB/s	462 B/s
🟢 TrueNAS-Scale 🟢	VM	129	3d 14h	1% (0.0/2 cores)	67% (5.3 GB/8.0 GB)	-	0 B/s	134 KB/s	2 KB/s	20 B/s
🟢 elastic	VM	116	2d 22h	8% (0.3/4 cores)	94% (17.0 GB/18.2 GB)	72% (1016.8 GB/1.4 TB)	200 B/s	10 MB/s	278 KB/s	49 KB/s
🟢 bentopdf 🟢🟢	LXC	224	2d 11h	0% (0.0/1 cores)	1% (36.4 MB/4.0 GB)	65% (2.5 GB/3.9 GB)	0 B/s	0 B/s	76 B/s	11 B/s
🟢 redhat-server	VM	112	1d 20h	2% (0.2/8 cores)	91% (8.9 GB/9.8 GB)	50% (47.1 GB/93.9 GB)	0 B/s	21 KB/s	10 KB/s	34 KB/s
🟢 redhat10	VM	118	1d 20h	2% (0.2/8 cores)	90% (8.6 GB/9.6 GB)	50% (45.7 GB/92.0 GB)	199 B/s	120 KB/s	9 KB/s	35 KB/s
🟢 Win2025Server	VM	126	21h 9m	4% (0.2/4 cores)	63% (4.3 GB/6.9 GB)	-	0 B/s	499 B/s	4 KB/s	33 B/s
🟢 wazuh 🟢🟢🟢 +2	LXC	214	13h 19m	1% (0.1/4 cores)	64% (2.6 GB/4.0 GB)	31% (35.4 GB/115.6 GB)	199 B/s	121 KB/s	79 KB/s	8 KB/s
🟢 rootca 🟢🟢🟢	LXC	250	-	0% (0.0/1 cores)	0% (0 B/512.0 MB)	0% (0 B/8.0 GB)	-	-	-	-
🟢 kms-iso 🟢🟢🟢	LXC	222	-	0% (0.0/1 cores)	0% (0 B/512.0 MB)	0% (0 B/8.0 GB)	-	-	-	-
🟢 debian-Extlan 🟢🟢🟢	LXC	216	-	0% (0.0/2 cores)	0% (0 B/2.0 GB)	0% (0 B/8.0 GB)	-	-	-	-
🟢 Ubuntu-pfS 🟢🟢🟢	LXC	212	-	0% (0.0/1 cores)	0% (0 B/512.0 MB)	0% (0 B/32.0 GB)	-	-	-	-
🟢 ubuntu-pfS2 🟢🟢🟢	LXC	211	-	0% (0.0/1 cores)	0% (0 B/512.0 MB)	0% (0 B/8.0 GB)	-	-	-	-

Backup dashboard

Reports on the various backup methods. The first screenshot provides details on the snapshot status and the second screenshot outlines the PBS status and history of backups.



Alert Dashboard and Notifications

Summary of current alerts, configured thresholds, notifications, schedule and alert history.

Pulse

Connected

Logout

Main

Storage

Backups

Alerts

Settings

Alert Configuration

Configure monitoring thresholds and notification settings

Overview

Thresholds

Notifications

Schedule

History

Active Alerts

6

Acknowledged

0

Last 24 Hours

6

Guest Overrides

0

Active Alerts

Select All

Jellyfin-Ubuntu (disk)

Container disk at 96.4%

Started: 8/29/2025, 5:06:17 PM

Nas (usage)

Storage at 85.5%

Started: 8/29/2025, 5:06:18 PM

homeassistos (memory)

VM memory at 97.4%

Started: 8/29/2025, 5:06:17 PM

UbuntuVM1 (memory)

VM memory at 95.5%

Started: 8/29/2025, 5:06:17 PM

Dockervm2 (memory)

VM memory at 94.2%

Started: 8/29/2025, 5:06:17 PM

splunk (memory)

VM memory at 97.6%

Started: 8/29/2025, 5:06:17 PM

[Pulse Alert] 21 Warning alerts

Delete

Archive

Report

Move to

Zoom

[Pulse Alert] 21 Warning alerts

shad0w1t1a6@gmail.com

To: You

Fri 1/9/2026 9:18 AM

Pulse Monitoring Alert Summary

21 New Alerts

21

Warning

RESOURCE	LEVEL	VALUE	DURATION
Kali-RedTeam	WARNING		
Ubuntu-pf1AN	WARNING		
Win11home	WARNING		
homeassistos	WARNING		
CAINE	WARNING		
eve-ng	WARNING		
cisco-vswitch	WARNING		
macos-tahoe	WARNING		
unused	WARNING		
ParrotOS7	WARNING		
win11pro2	WARNING		

[Pulse Alert] 1 Critical alert

[Pulse Alert] 1 Critical alert

shad0w1t1a6@gmail.com

To: You

Fri 1/9/2026 10:10 AM

Pulse Monitoring Alert Summary

1 New Alert

1

Critical

RESOURCE	LEVEL	VALUE	DURATION
win2022server	CRITICAL	110.7% of 80%	30 seconds ago

This is an automated notification from Pulse Monitoring
View alerts and configure settings in your Pulse dashboard

Network Security Monitoring Sensor – Zeek and ntopng

Deployment Overview

The NSM host is a dedicated Ubuntu-based sensor VM running on Proxmox, purpose-built for deep packet inspection, network flow collection, and protocol metadata extraction. It operates as a passive, observe-only node — no traffic is routed through it. Zeek provides full protocol metadata logging across all monitored segments; ntopng provides flow-level visibility via NetFlow v9 ingestion from Cisco infrastructure. Log output feeds both the Elastic stack for long-term retention and Brim/Zui for local forensic analysis.

Security Impact

- Full protocol metadata across all monitored network segments (DNS, HTTP, SSL/TLS, SSH, SMB, RDP, DCE-RPC)
- NetFlow-based east-west and north-south traffic visibility
- File hashing and Notice/Intel/Weird framework detections from Zeek
- Passive-only design eliminates sensor as an attack vector or pivot point
- SOC-grade log output available for correlation, alerting, and forensic review

Deployment Rationale

General infrastructure monitoring tools provide host and service metrics but no network-layer visibility. The NSM host fills this gap with dedicated packet capture and flow analysis capability, mirroring enterprise SOC sensor architectures. Zeek and ntopng together provide both metadata-level detail and flow-level aggregation, enabling detection use cases that SIEM and EDR alone cannot cover.

Architecture Principles Alignment

- **Defense in Depth:** Network metadata layer complements host-based EDR (Wazuh), IDS/IPS (Suricata/Snort), and firewall logging; detects lateral movement and protocol anomalies not visible at the endpoint
- **Secure by Design:** Capture NICs are isolated from the management interface; the host never routes or forwards traffic
- **Zero Trust:** All traffic is treated as potentially malicious; full metadata logging supports continuous verification and retroactive investigation

Zeek — Protocol Metadata Engine

Zeek runs in a cluster layout to maximize throughput across three capture interfaces. All log output is JSON-formatted for direct ingestion by Elastic and Brim/Zui.

Cluster Layout

Role	Host	Interface	AF_PACKET Fanout ID
Manager	localhost	—	—
Proxy	localhost	—	—
worker-1	localhost	enp6s19	20
worker-2	localhost	enp6s20	21
worker-3	localhost	enp6s21	22

Monitored Networks

Subnet	Label
192.168.1.0/24	prod_lan

192.168.100.0/24 lab_lan1

192.168.200.0/24 lab_lan2

Enabled Frameworks and Analyzers

- Protocol analyzers: DNS, HTTP, SSL/TLS, SSH, SMB, RDP, DCE-RPC
- Notice framework
- Intel framework
- Weird framework
- File hashing

Directory Layout

/opt/monitoring/zeek/
 site/ # local.zeek, JSON logging policy, tuning
 logs/ # timestamped logs + current/ symlink
 spool/ # cluster runtime state

Configuration

node.cfg

```
[zeek]
Type=manager
host=localhost

[proxy]
type=proxy
host=localhost

[worker-1]
type=worker
host=localhost
interface=enp6s19
lb_method=af_packet
lb_procs=1
af_packet_fanout_id=20

[worker-2]
type=worker
host=localhost
interface=enp6s20
lb_method=af_packet
lb_procs=1
af_packet_fanout_id=21

[worker-3]
type=worker
host=localhost
interface=enp6s21
lb_method=af_packet
lb_procs=1
af_packet_fanout_id=22
```

Configuration
networks.cfg

```
192.168.1.0/24  prod_lan
192.168.100.0/24 lab_lan1
192.168.200.0/24 lab_lan2
```

Configuration
local.zeek
(relevant additions)

```
@load policy/tuning/json-logs
@load json-streaming-logs

redef JSONStreaming::disable_default_logs = T;
redef LogAscii::json_timestamps = JSON::TS_ISO8601;
```

Sample Output
conn.log (JSON)

```
{
  "_path": "conn",
  "_write_ts": "2026-03-03T16:26:18.414917Z",
  "ts": "2026-03-03T16:25:48.414242Z",
  "uid": "C2eBK5DVFxx8BBe81",
  "id.orig_h": "192.168.3.31",
  "id.orig_p": 137,
  "id.resp_h": "192.168.3.255",
  "id.resp_p": 137,
  "proto": "udp",
  "service": "dns",
  "duration": 13.967844,
  "orig_bytes": 450,
  "resp_bytes": 0,
  "conn_state": "S0",
  "local_orig": true,
  "local_resp": true,
  "missed_bytes": 0,
  "history": "D",
  "orig_pkts": 9,
  "orig_ip_bytes": 702,
  "resp_pkts": 0,
  "resp_ip_bytes": 0,
  "ip_proto": 17
}
```

ntopng — Flow Visibility

ntopng ingests NetFlow v9 exports from pfSense, OPNsense and Cisco infrastructure (r1, r2, r3) and provides real-time and historical flow analysis via a web UI. It complements Zeek's per-connection metadata with aggregated flow-level visibility across the network.

Two nProbe instances have been configured to avoid the community edition limitation of 5,000 flows. Nprobe1 collects live flows from the Cisco devices while nProbe2 collects from the firewalls.

**Configuration
ntopng.conf**

```
--http-port=3000
--local-networks="192.168.3.0/24,192.168.100.0/24,192.168.200.0/24"
--data-dir=/opt/monitoring/ntopng/data
--community
-i=tcp://127.0.0.1:5556
-i=tcp://127.0.0.1:5557
-i=enp6s18
-i=enp6s19
-i=enp6s20
-i=enp6s21
```

**Configuration
nprobe.conf**

```
--ntopng=zmq://127.0.0.1:5556
--collector-port=2055
-n=none
-i=none
-T="@NTOPNG@"
--lifetime-timeout 60
--idle-timeout 15
```

**Configuration
nprobe-
firewalls.conf**

```
--ntopng=zmq://127.0.0.1:5557
--collector-port=2056
-n=none
-i=none
-T="@NTOPNG@"
--lifetime-timeout 60
--idle-timeout 15
```

NetFlow Export — Cisco Configuration

Cisco vIOS routers export Flexible NetFlow v9 to the NSM host:

- Export target: 192.168.1.48 UDP/2055
- Source: stable SVI or loopback interface
- Active/inactive timeouts tuned for ntopng collector

Sample Net Flow Configuration

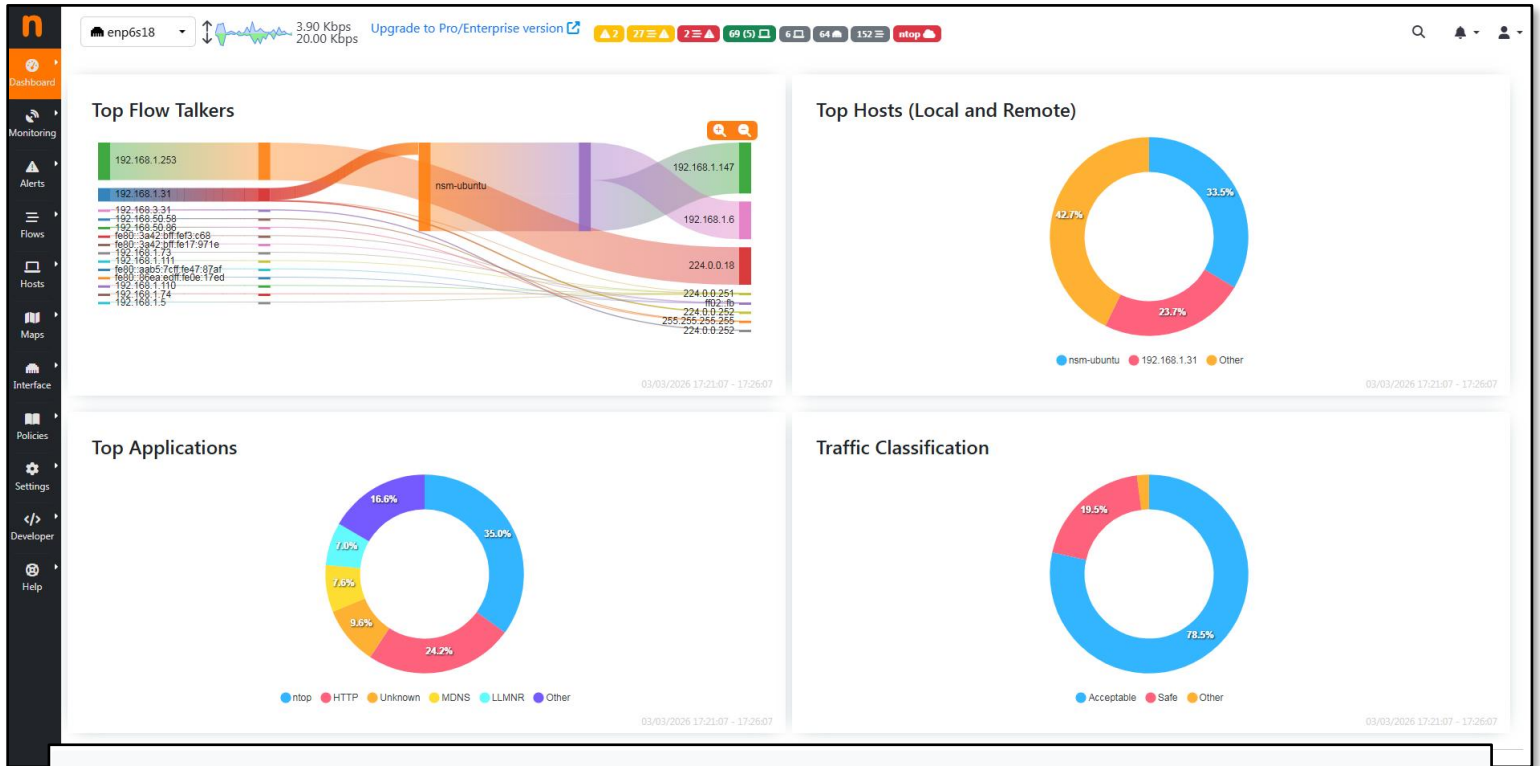
```
flow record ntop-record
description NetFlow record for ntopng
match ipv4 source address
match ipv4 destination address
match transport source-port
match transport destination-port
match interface input
match interface output
match ipv4 tos
match flow direction
match ipv4 protocol
collect counter bytes long
collect counter packets long
collect timestamp sys-uptime first
collect timestamp sys-uptime last

flow exporter ntop-exporter
description Export to ntopng
destination 192.168.1.48
source Loopback1
transport udp 2055
template data timeout 60

flow monitor ntop-monitor
description Monitor for ntopng
exporter ntop-exporter
cache timeout active 60
record ntop-record

interface GigabitEthernet0/1-3
ip flow monitor ntop-monitor input
ip flow monitor ntop-monitor output
```


Interface – Prod_LAN captures



Live Flows Aggregation									
Host	Protocol	Application	Status	Traffic Type	VLAN	Host Pools	Networks		
All	All	All	All	All	All	All	All	Reset	50
Actions	Last Seen	Duration	Protocol	Score	Flow	Actual Thpt	Total Bytes		
	< 1 sec	19:26:23	ICMP		nsm-ubuntu ↔ Cisco r3	885.12 bps	6.99 MB		
	< 1 sec	19:26:23	UDP:NetFlow		Cisco r3 : 50502 ↔ nsm-ubuntu : 2055	735.82 bps	6.03 MB		
	< 1 sec	21:53:48	VRRP		192.168.1.253 ↔ vrrp.mcast.net	559.99 bps	5.21 MB		
	00:03	17:09	TCP:NetBIOS.SMBv23 ...		192.168.1.31 : 63932 ↔ nsm-ubuntu : 445	21.33 Kbps	4.40 MB		
	00:15	21:53:29	UDP:MDNS		192.168.3.31@30 : 5353 ↔ mdns.mcast.net@30 : 5353	789.37 bps	4.14 MB		
	00:15	21:53:29	UDP:MDNS		192.168.1.31 : 5353 ↔ mdns.mcast.net : 5353	746.70 bps	3.92 MB		
	00:02	05:13:48	UDP:NetFlow		Cisco r1 : 62251 ↔ nsm-ubuntu : 2055	5.36 Kbps	2.28 MB		
	00:02	05:13:48	ICMP		nsm-ubuntu ↔ Cisco r1	3.15 Kbps	1.49 MB		
	00:14	21:53:34	UDP:NetBIOS		192.168.3.31@30 : 137 ↔ 192.168.3.255@30 : 137	768.22 bps	1.38 MB		
	00:14	21:53:34	UDP:NetBIOS		192.168.1.31 : 137 ↔ 192.168.1.255 : 137	736.22 bps	1.33 MB		
	00:01	21:53:44	UDP:SSDP		192.168.3.31@30 : 63086 ↔ 192.168.3.255@30 : 32414	178.66 bps	1.01 MB		
	00:01	21:53:44	UDP:SSDP		192.168.3.31@30 : 63084 ↔ 192.168.3.255@30 : 32412	178.66 bps	1.01 MB		
	00:01	21:53:44	UDP:SSDP		192.168.1.31 : 63085 ↔ 192.168.1.255 : 32414	168.00 bps	968.13 KB		
	00:01	21:53:44	UDP:SSDP		192.168.1.31 : 63083 ↔ 192.168.1.255 : 32412	168.00 bps	968.13 KB		
	00:13	21:53:31	IGMP		192.168.1.142 ↔ igmp.mcast.net	432.02 bps	593.96 KB		
	00:11	09:30:46	UDP:SSDP		Google-Home-Mini@50 : 36885 ↔ ff02::c@50 : 1900	943.96 bps	591.84 KB		
	00:10	08:52:44	UDP:SSDP		Google-Home-Mini@50 : 37287 ↔ ff02::c@50 : 1900	943.96 bps	552.26 KB		
	00:01	08:42:20	UDP:SSDP		Google-Home-Mini@50 : 53132 ↔ ff02::c@50 : 1900	943.98 bps	541.72 KB		
	00:16	21:53:24	IGMP		192.168.1.152 ↔ igmp.mcast.net	143.94 bps	406.89 KB		
	00:13	21:53:29	IGMP		hyper-v-2019 ↔ igmp.mcast.net	143.99 bps	406.30 KB		

Host: Cisco r3

TrafficPacketsPortsPeersICMPAppsDNS

MAC Address	ProxmoxServe_79:D3:39		Networking
IP Address	192.168.1.147 [192.168.1.0/24]		Host Pool: Default
Name	Cisco r3		
Active Monitoring	Add ICMP Monitor +		
Behavioural Counter Anomalies	1		
First / Last Seen	02/03/2026 17:00:12 [19:28:04 ago]	03/03/2026 12:28:16 [< 1 sec ago]	
Sent vs Received Traffic Breakdown	Sent Recv		
Traffic Sent / Received	37,149 Pkts / 6.1 MB ↑	37,485 Pkts / 7.1 MB ↑	
CVEs	Add Host to Vulnerability Scan List		
	As Client	As Server	
Flows: Active / Total / Alerted / Port Unreach	1 — / 77 — / 76 — / 76 —	1 — / 305 — / 76 — / 1 —	
Total Flows with Blacklisted Hosts	0 —	0 —	
Total Unidirectional TCP/UDP Flows	0 (0.0 %) —	76 (24.9 %) —	
Peers: Active	1 —	1 —	
TCP/UDP Unresponsive Flows (Peer IP and Server Port)	0 —	1 —	
Contacted Servers	DNS: 0 / SMTP: 0 / POP: 0 / IMAP: 0 / NTP: 0		
Reset Host Stats	Reset Host Stats Reset Blacklisted Hosts Stats		
Download	JSON		1 min Filter (BPF) pcap Download

nProbe1 – Cisco Device Flows:

nProbe for Cisco Devices

0 bps0 bps

24 (U)4442stop

DashboardAlertsFlowsHostsMapsInterfacePoliciesSettingsDeveloperHelp

Top Flow Talkers

05/03/2026 16:30:12 - 16:35:12

Top Hosts (Local and Remote)

05/03/2026 16:30:12 - 16:35:12

Top Applications

05/03/2026 16:30:12 - 16:35:12

Traffic Classification

05/03/2026 16:30:12 - 16:35:12

ntopng Community v6.7.260305 (Ubuntu 24.04.4 LTS)

© 1998-26 - ntop

16:35:18 +0000 UTC | Uptime: 35:25

Live Flows Aggregation						
Host	Protocol	Application	Status	Traffic Type	Host Pools	Networks
All ▾	All ▾	All ▾	All ▾	All ▾	All ▾	All ▾
Reset						
Actions	Last Seen	Duration	Protocol	Score	Flow	
	00:23	01:00	UDP:NetFlow Guess		Cisco R3 : 50502 nsm-ubuntu : 2055	
	00:43	00:32	UDP:NetFlow Guess		Cisco R2 : 64793 nsm-ubuntu : 2055	
	00:22	00:56	UDP:MDNS Guess		192.168.3.31 : 5353 mdns.mcast.net : 5353	
	00:22	00:56	UDP:MDNS Guess		192.168.1.31 : 5353 mdns.mcast.net : 5353	
	00:07	00:39	OSPF		10.30.0.3 ospf-all.mcast.net	
	00:21	00:32	UDP:NetBIOS Guess		192.168.3.31 : 137 192.168.3.255 : 137	
	00:21	00:33	UDP:NetBIOS Guess		192.168.1.31 : 137 192.168.1.255 : 137	
	00:08	01:13	UDP:Unknown Guess		192.168.3.31 : 58480 192.168.3.255 : 32414	
	00:05	00:04	UDP:Unknown Guess		192.168.3.31 : 58478 192.168.3.255 : 32412	
	00:10	00:19	UDP:Unknown Guess		192.168.1.31 : 58479 192.168.1.255 : 32414	
	00:05	00:24	UDP:Unknown Guess		192.168.1.31 : 58477 192.168.1.255 : 32412	
	00:19	00:01	UDP:SNMP Guess		Cisco R2 : 161 192.168.1.126 : 39732	
	00:45	00:01	OSPF		10.30.0.2 ospf-all.mcast.net	
	00:26	00:01	OSPF		10.30.0.1 ospf-all.mcast.net	
	00:03	00:01	UDP:SNMP Guess		192.168.1.126 : 35330 Cisco R2 : 161	
	00:19	00:01	UDP:SNMP Guess		Cisco R2 : 161 192.168.1.126 : 35988	
	00:03	00:01	UDP:SNMP Guess		192.168.1.126 : 45122 Cisco R2 : 161	
	00:03	00:01	UDP:SNMP Guess		192.168.1.126 : 43146 Cisco R2 : 161	
	00:03	00:01	UDP:SNMP Guess		192.168.1.126 : 52608 Cisco R2 : 161	
	00:03	00:01	UDP:SNMP Guess		192.168.1.126 : 57541 Cisco R2 : 161	
	00:03	00:01	UDP:SNMP Guess		192.168.1.126 : 49628 Cisco R2 : 161	
	00:03	00:01	UDP:SNMP Guess		192.168.1.126 : 42549 Cisco R2 : 161	
	00:03	00:01	UDP:SNMP Guess		192.168.1.126 : 52495 Cisco R2 : 161	
	00:03	00:01	UDP:SNMP Guess		192.168.1.126 : 55822 Cisco R2 : 161	

Brim/ZUI Queries

[illegible]

zeekwiresharkQuery Session

← →Untitled

from wireshark

1count() by query | sort -r count

No data.

2 Shapes — Filter or [fuse](#) to view results as one shape.

> {query: "functional.events.data.microsoft.com", count: 134 (uint64)}

> {query: "wpad", count: 107 (uint64)}

> {query: "WPAD", count: 84 (uint64)}

> {query: "_googlecast._tcp.local", count: 51 (uint64)}

> {query: "d.docs.live.net", count: 18 (uint64)}

> {query: "qbt.home.com", count: 18 (uint64)}

> {query: "*", count: 17 (uint64)}

> {query: "bravia-4k-gb-5a500ac9bc387bae6cc273528720d693._googlecast._tcp.local", count: 15 (uint64)}

> {query: "www.1337x.to", count: 14 (uint64)}

> {query: "_googlezone._tcp.local", count: 14 (uint64)}

> {query: "_spotify-connect._tcp.local", count: 13 (uint64)}

> {query: "http-intake.logs.datadoghq.com", count: 12 (uint64)}

> {query: "e1586b46-be5f-f19b-6dca-034859160c85._googlezone._tcp.local", count: 8 (uint64)}

> {query: "65fa36c8-d032-68b9-b139-425e3a3c1912._googlezone._tcp.local", count: 6 (uint64)}

> {query: "143.141.67.172.in-addr.arpa", count: 6 (uint64)}

> {query: "41.94.18.104.in-addr.arpa", count: 6 (uint64)}

> {query: "outlook.office.com", count: 6 (uint64)}

> {query: "gue1-spclient.spotify.com", count: 6 (uint64)}

> {query: "api.honeycomb.io", count: 6 (uint64)}

> {query: "6000f490-436f-8ede-c156-5bd541a70b17._googlezone._tcp.local", count: 6 (uint64)}

> {query: "google-home-mini-65fa36c8d03268b9b139425e3a3c1912._googlecast._tcp.local", count: 4 (uint64)}

> {query: "remotedesktop-pa.googleapis.com", count: 3 (uint64)}

> {query: "google-cast-group-dc5ba7c390304fff9fce8aebef5db50-1._googlecast._tcp.local", count: 3 (uint64)}

> {query: "family room._airplay._tcp.local", count: 3 (uint64)}

> {query: "aepxlg.adobe.com", count: 3 (uint64)}

> {query: "challenges.cloudflare.com", count: 3 (uint64)}

> {query: "turing-writingassistance.edge.microsoft.com", count: 2 (uint64)}

> {query: "editor.svc.cloud.microsoft", count: 2 (uint64)}

> {query: "30a4da4e-8019-2b6a-7665-71398bd2c811._googlezone._tcp.local", count: 2 (uint64)}

> {query: "family ultra._airplay._tcp.local", count: 2 (uint64)}

> {query: "torimage.com", count: 2 (uint64)}

> {query: "bedroom tv._androidtvremote2._tcp.local", count: 2 (uint64)}

> {query: "edge.microsoft.com", count: 2 (uint64)}

> {query: "signaler-pa.clients6.google.com", count: 2 (uint64)}

> {query: "un1-chat.heroesofhistorygame.com", count: 2 (uint64)}

> {query: "bcfb44bc-1662-d042-d2aa-934ff72bf193._googlezone._tcp.local", count: 2 (uint64)}

NetAlertx Network Visibility & Asset Intelligence Framework.

Deployment Overview



NetAlertx operates as a nested Docker container on a LXC host using host networking mode for comprehensive network discovery. The deployment provides continuous asset inventory, device profiling, and network topology mapping across production and lab segments (Prod_LAN, Lab_LAN1, Lab_LAN2, Ext_LAN). The container leverages multiple NIC interfaces on the host for passive and active scanning while maintaining isolation from sensitive ISO_LAN segments to prevent credential exposure or unauthorized enumeration.

Security Impact

- Real-time asset visibility identifies rogue devices and unauthorized network connections
- Automated device profiling detects configuration drift and shadow IT deployments
- Network topology mapping reveals unexpected lateral movement paths
- MAC address tracking correlates device identity across DHCP lease changes
- Alert generation on new device detection enables rapid security response

Deployment Rationale

Enterprise SOC's require accurate asset inventories for vulnerability management, incident response, and compliance reporting. NetAlertx provides continuous discovery without agent deployment, identifying IoT devices, network appliances, and ephemeral containers that evade traditional endpoint management. Deployment on the WAF host consolidates network security functions while maintaining separation from production workloads.

Architecture Principles Alignment

- **Defense in Depth:** Asset visibility layer complements firewall ACLs and IDS/IPS by identifying security gaps in network segmentation
- **Secure by Design:** Passive discovery minimizes network impact; isolated from sensitive subnets to prevent reconnaissance against high-value targets
- **Zero Trust:** Continuous device enumeration validates network access aligns with authorized asset inventory; detects policy violations

High-Level Configuration

- **Deployment Model:** Docker container with host networking (--network host)
- **Discovery Methods:** ARP scanning, DHCP lease monitoring, mDNS/SSDP enumeration, SNMP polling
- **Monitored Networks:** 192.168.1.0/24 (Prod_LAN), 192.168.100.0/24 (Lab_LAN1), 192.168.200.0/24 (Lab_LAN2), 192.168.2.0/24 (Ext_LAN)
- **Exclusions:** 10.20.0.0/24 and 192.168.3.0/24 (ISO_LAN segments) - intentionally excluded from scanning
- **Alert Triggers:** New device detection, unknown MAC addresses, duplicate IP assignments, offline device recovery
- **Integration:** Webhook notifications to SOC alerting pipeline; asset data exported for vulnerability correlation

Connected Devices

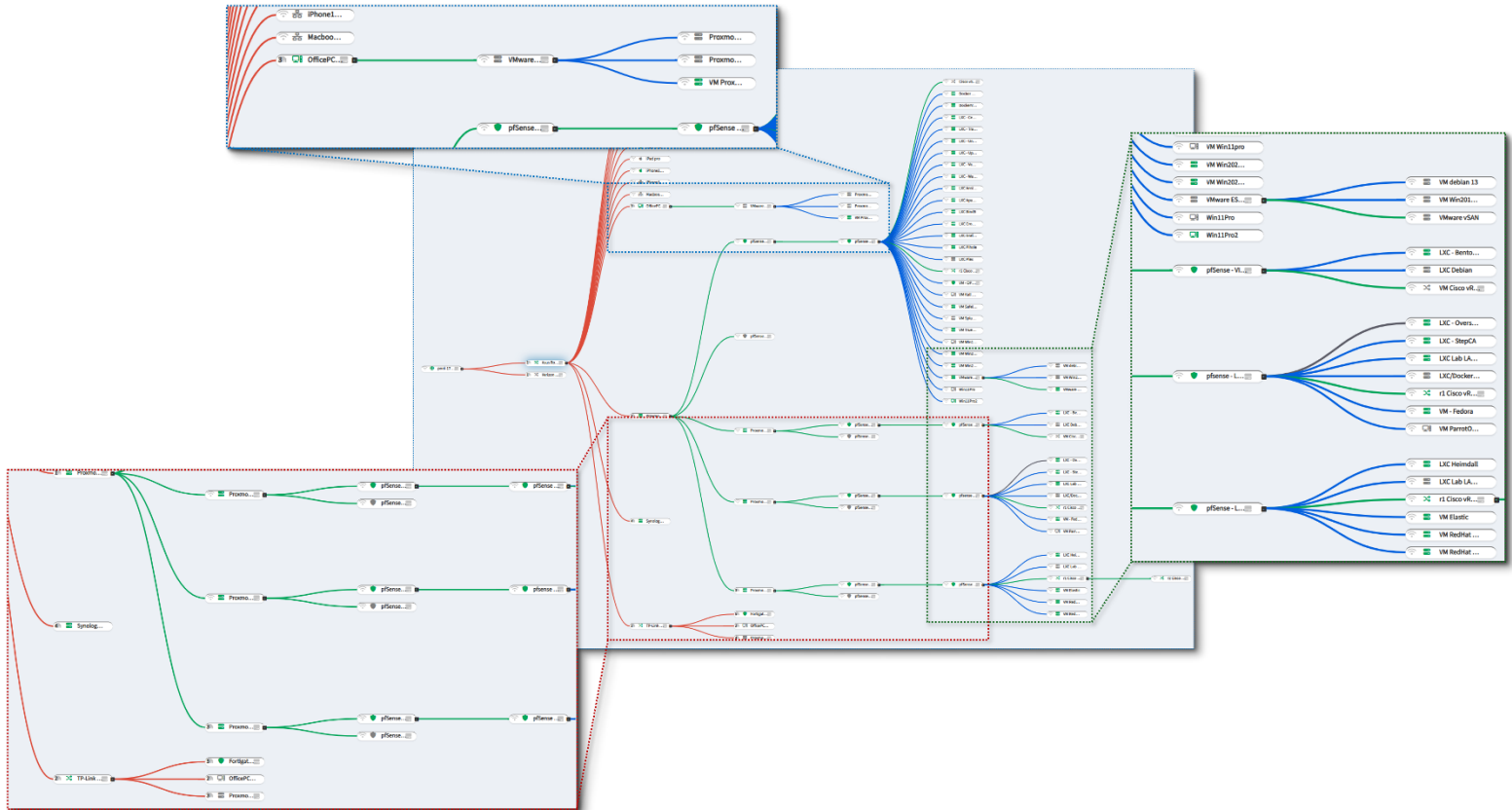
Virtual
Machines

Icon	Name	Type	Last IP	Status	Full MAC
	LXC - Vaultwarden	VM	192.168.1.4	New	bc:24:11:e6:51:36
	LXC Ansible	VM	192.168.1.25	New	bc:24:11:96:2a:dc
	LXC Crowdsec	VM	192.168.1.33	New	bc:24:11:78:26:21
	LXC - CentOS	VM	192.168.1.93	New	bc:24:11:dc:84:e4
	LXC Apache Webserver	VM	192.168.1.108	New	bc:24:11:f9:a0:8c
	DockerVM1 - Ubuntu	VM	192.168.1.126	New	bc:24:11:cf:5a:ad
	VM Win2025 Server	VM	192.168.1.142	Online	bc:24:11:83:4e:b5
	VM Win2022 Server - DC01	VM	192.168.1.152	Online	bc:24:11:ff:0c:24
	Docker VM2 - Debian	VM	192.168.1.166	New	bc:24:11:0a:69:cc
	LXC - Uptime kuma	VM	192.168.1.181	New	bc:24:11:0b:98:75
	LXC - Wazuh	VM	192.168.1.219	New	bc:24:11:ca:5a:0a
	VM Proxmox Backup Server	VM	192.168.1.243	New	00:0c:29:31:06:f7
	LXC Grafana	VM	192.168.1.246	New	bc:24:11:e6:bd:e1
	LXC - Traefik	VM	192.168.1.247	New	bc:24:11:81:e3:8e
	LXC Pi-hole	VM	192.168.1.250	New	bc:24:11:48:96:ba
	LXC Bind9	VM	192.168.1.251	New	bc:24:11:a8:de:cf
	LXC - Unbound	VM	192.168.1.252	New	bc:24:11:93:48:bb
	VM Safeline Docker	VM	192.168.1.89	New	bc:24:11:4e:f9:ba
	VM - Fedora	VM	192.168.100.5	Online	bc:24:11:e1:88:84
	LXC - StepCA	VM	192.168.100.51	New	bc:24:11:8e:36:42
	LXC - BentoPDF	VM	192.168.2.12	New	bc:24:11:4f:70:11
	LXC Heimdall	VM	192.168.200.7	New	bc:24:11:c3:57:70
	VM Elastic	VM	192.168.200.8	New	bc:24:11:a2:d0:af
	VM RedHat K3s-Worker	VM	192.168.200.21	New	bc:24:11:6f:3e:f6
	VM RedHat K3s-Control	VM	192.168.200.22	New	bc:24:11:b7:dd:d6
	LXC Lab LAN Host1	VM	192.168.100.4	Online	bc:24:11:59:31:45
	LXC Ovrsr	VM	192.168.100.15	Online	bc:24:11:29:b0:d0

Network Devices

Icon	Name	Type	Last IP	Status	Full MAC
	Proxmox Host Server - LAN2 Interface	Hypervisor	192.168.200.178 [↗]	 Online	7a:c6:fb:ec:ca:d9
	r1 Cisco vRouter	Router	192.168.200.6 [↗]	 Online	bc:24:11:6c:c6:bc
	pfSense-B - LAN2 Interface	Firewall	192.168.200.3 [↗]	 New	bc:24:11:54:42:4a
	pfSense-A - LAN2 Interface	Firewall	192.168.200.2 [↗]	 New	bc:24:11:ad:2d:de
	pfSense - LAN2 VIP	Firewall	192.168.200.1 [↗]	 New	00:00:5e:00:01:03
	Proxmox Host Server - LAN1 Interface	Hypervisor	192.168.100.10 [↗]	 Online	96:6e:b5:42:8e:2a
	r1 Cisco vRouter - Lab LAN Interface	Router	192.168.100.6 [↗]	 Online	bc:24:11:69:c5:e9
	pfSense-B - LAN1 Interface	Firewall	192.168.100.3 [↗]	 New	bc:24:11:06:cb:9f
	pfSense-A - LAN1 Interface	Firewall	192.168.100.2 [↗]	 New	bc:24:11:5d:66:95
	pfSense - LAN1 VIP	Firewall	192.168.100.1 [↗]	 New	00:00:5e:00:01:02
	Proxmox Host Server - VLAN3	Hypervisor	192.168.3.178 [↗]	 Offline	98:b7:85:21:c6:bf
	Proxmox Host Server - Ext LAN Interface	Hypervisor	192.168.2.178 [↗]	 Online	ca:5b:d1:00:50:ac
	r2 Cisco vRouter Ext LAN Interface	Router	192.168.2.9 [↗]	 Online	bc:24:11:83:ae:a4
	VM Cisco vRouter Ext LAN Interface	Router	192.168.2.6 [↗]	 New	bc:24:11:ca:33:7e
	pfSense-B - Ext LAN Interface	Firewall	192.168.2.3 [↗]	 New	bc:24:11:78:bd:4a
	pfSense-A - Ext LAN Interface	Firewall	192.168.2.2 [↗]	 New	bc:24:11:5b:9f:f7
	pfSense - VIP Ext LAN	Firewall	192.168.2.1 [↗]	 New	00:00:5e:00:01:04
	pfSense - VIP Prod LAN	Firewall	192.168.1.254 [↗]	 New	00:00:5e:00:01:01
	pfSense-A - Prod LAN interface	Firewall	192.168.1.253 [↗]	 New	bc:24:11:4e:2c:89
	pfSense-B Prod LAN Interface	Firewall	192.168.1.249 [↗]	 New	bc:24:11:2d:e9:bf

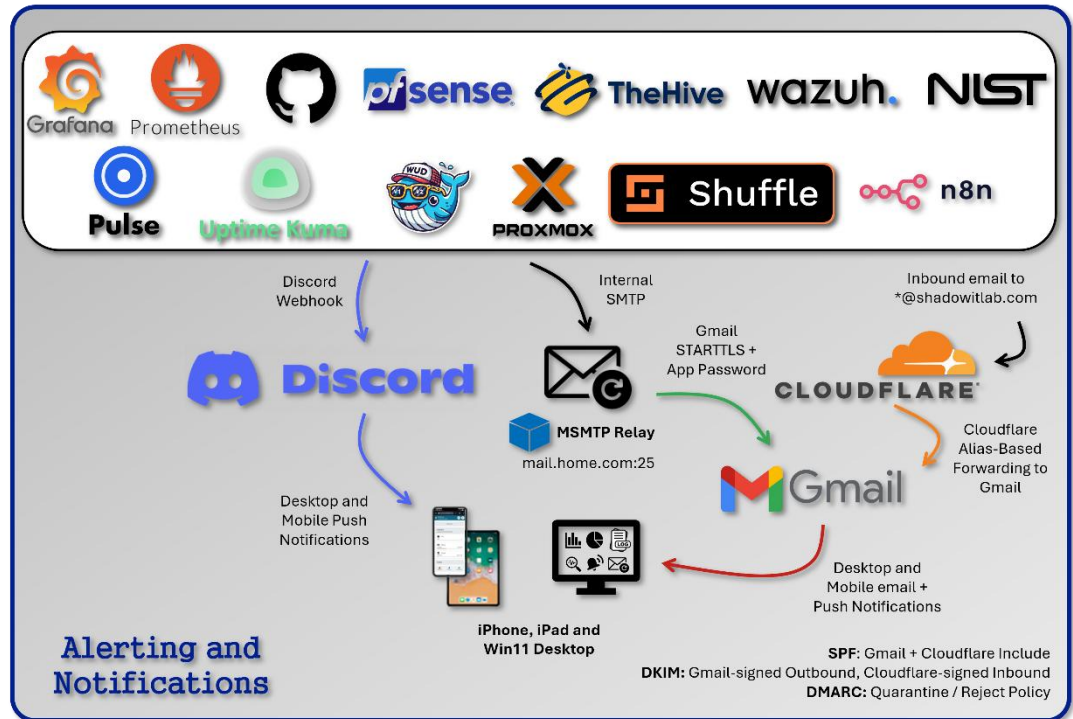
Network Connectivity Map



Alerting and Notification Architecture

Deployment Overview

The alerting architecture ensures real-time visibility into operational and security events across the entire lab environment. Alerts from monitoring tools, SIEM platforms, EDR agents, and infrastructure services are routed through multiple redundant channels (Discord, SMTP relay, and Cloudflare email routing). This multi-path design ensures no critical alert is ever missed and enables granular triage through dedicated service-specific channels.



Security Impact

- Immediate notification of outages, anomalies, or security events
- Redundant channels prevent alert loss due to single-system failure
- Segmented channels reduce noise and improve triage efficiency

Deployment Rationale

Modern SOC/NOC operations rely on multi-channel alerting to ensure rapid response. This architecture demonstrates proficiency with webhook-based alerting, SMTP relays, email routing, and real-time collaboration platforms.

Architecture Principles Alignment

- **Defense in Depth:** Multiple alerting paths ensure resilience
- **Secure by Design:** TLS-encrypted SMTP; restricted webhook endpoints
- **Zero Trust:** Alerts validated and routed per-service; no implicit trust in any channel

Discord Private Server – Centralized Notification Hub

Deployment Overview



A private Discord server acts as the real-time alerting hub for the entire environment. Each monitored service has its own dedicated channel, enabling noise isolation, targeted triage, and clear operational separation.

Security Impact

- Instant visibility into outages, anomalies, and security alerts
- Channel segmentation prevents alert overload
- Provides audit trails for incident response

Deployment Rationale

Discord offers low-latency notifications, webhook integration, and structured channel organization—mirroring enterprise chat-ops workflows.

Architecture Principles Alignment

- **Defense in Depth:** Secondary alerting path alongside email
- **Secure by Design:** Private server; restricted webhook tokens
- **Zero Trust:** No alert is trusted without validation; all events logged

Notification Flow

[Monitoring Source] → [Alert Trigger] → [Webhook or Script] → [Discord Channel] → [Push Notification]

- Monitoring tools detect anomalies or threshold breaches.
- Alerts are triggered via native webhook integrations or custom scripts.
- Messages are sent to service-specific Discord channels using Discord's webhook API.
- Discord's push notification system delivers alerts to:
 - Windows 11 desktop app
 - iPad and iPhone mobile apps

Discord Channel Structure

Channel Name	Source Tool	Alert Type
#proxmox	Proxmox VE	Node status, backup failures
#grafana	Grafana	Dashboard alerts, data source issues
#splunk	Splunk	Security events, log anomalies
#wazuh	Wazuh	Host-based intrusion alerts
#pulse	Pulse	OpenSearch monitor triggers
#uptime-kuma	Uptime Kuma	Service downtime, latency spikes
#openvas	OpenVAS	Vulnerability scan results
#pfsense	pfSense	Firewall events, VPN status
#prometheus	Prometheus	Node exporter, resource thresholds
#authentik	Authentik	Auth flow errors/alarms, auth failures
#Windowsupdates	Windows VMs	Python script completion

Each channel is configured to receive only relevant alerts, reducing noise and improving signal clarity.

SMTP Relay – Gmail-Backed Email Alerting

Gmail relay to dedicated email address, shad0w1t1a6@gmail.com.



Deployment Overview

An msmtplib container provides a secure SMTP relay to Gmail using STARTTLS and application-specific passwords. Internal services send alerts via standard SMTP on port 25, and msmtplib handles encrypted delivery to the dedicated alert mailbox.

Configured services include:

Proxmox, pfSense, OPNsense, TheHive, Uptime Kuma, Wazuh, Grafana, n8n, Synology NAS.

Security Impact

- Provides a reliable, encrypted alerting channel
- Ensures alerts are preserved even if chat-ops channels fail
- Supports forensic review through email retention

Deployment Rationale

Email remains a universal, durable alerting mechanism. This relay demonstrates secure outbound email configuration and multi-service integration.

Architecture Principles Alignment

- **Defense in Depth:** Email complements Discord for redundancy
- **Secure by Design:** STARTTLS encryption; app-password authentication
- **Zero Trust:** All alerts validated and logged; no implicit trust in sender

Cloudflare Email Routing

Deployment Overview



Cloudflare Email Routing provides alias-based forwarding for individual services, enabling clean separation of alert sources and simplified identity management. Each service is assigned a unique alias for traceability.

Security Impact

- Prevents spoofing by enforcing domain-validated routing
- Enables per-service alert attribution
- Provides an additional layer of redundancy

Deployment Rationale

Cloudflare's routing service mirrors enterprise email aliasing strategies, improving traceability and reducing operational complexity.

Architecture Principles Alignment

- **Defense in Depth:** Adds routing redundancy and identity separation
- **Secure by Design:** Domain-validated forwarding; Cloudflare-managed security
- **Zero Trust:** Each alias treated as an independent identity; no implicit trust

Custom address	Action	Destination	Status
grafana@shadowitlab.com	Send to an email	shad0w1t1a6@gmail.com	 Active
n8n@shadowitlab.com	Send to an email	shad0w1t1a6@gmail.com	 Active
truenas@shadowitlab.com	Send to an email	shad0w1t1a6@gmail.com	 Active
prom@shadowitlab.com	Send to an email	shad0w1t1a6@gmail.com	 Active
uptime@shadowitlab.com	Send to an email	shad0w1t1a6@gmail.com	 Active
elastic@shadowitlab.com	Send to an email	shad0w1t1a6@gmail.com	 Active
pfsense@shadowitlab.com	Send to an email	shad0w1t1a6@gmail.com	 Active
wazuh@shadowitlab.com	Send to an email	shad0w1t1a6@gmail.com	 Active
proxmox@shadowitlab.com	Send to an email	shad0w1t1a6@gmail.com	 Active
thehive@shadowitlab.com	Send to an email	shad0w1t1a6@gmail.com	 Active
soc@shadowitlab.com	Send to an email	shad0w1t1a6@gmail.com	 Active
pnleone@shadowitlab.com			

Sender	Custom address	Received	Result
pnleone@outlook.com	proxmox@shadowitlab.com	4 minutes ago	Forwarded
shad0w1t1a6@gmail.com	proxmox@shadowitlab.com	7 minutes ago	Forwarded
shad0w1t1a6@gmail.com	pfsense@shadowitlab.com	10 minutes ago	Forwarded
shad0w1t1a6@gmail.com	pfsense@shadowitlab.com	12 minutes ago	Forwarded
shad0w1t1a6@gmail.com	pnleone@shadowitlab.com	8 hours ago	Forwarded

Security Controls Summary

Visibility and Response Security Controls

Control Framework:

Control Domain	Implementation	Coverage
Log Aggregation	"Splunk + Elastic centralized collection"	"All infrastructure"
Endpoint Detection	"Wazuh EDR on 12 hosts"	"Windows, Linux, MacOS"
Network Monitoring	"Suricata/Snort IDS on all interfaces"	"100% traffic inspection"
File Integrity Monitoring	"Wazuh FIM on critical paths"	"System files, configs"
Vulnerability Assessment	"Wazuh + OpenVAS automated scanning"	"All hosts"

Compliance Auditing	"Wazuh SCA policies (CIS, PCI-DSS)"	"Continuous assessment"
Alert Correlation	"Splunk SPL queries, Elastic detection rules"	"Multi-source correlation"
Incident Response	"Automated containment, Discord alerting"	"Real-time response"
Audit Logging	"90-day retention in SIEM"	"Full audit trail"
Performance Monitoring	"Prometheus + Grafana dashboards"	"All infrastructure"

Security Event Pipeline:

- Collection: Agents/forwarders collect logs from sources
- Transport: Encrypted TLS/SSL to SIEM platforms
- Parsing: Field extraction and normalization
- Enrichment: GeoIP, threat intel, asset context
- Correlation: Multi-source event correlation
- Detection: Rule-based and ML anomaly detection
- Alerting: Discord webhooks + email notifications
- Response: Automated containment actions
- Investigation: Search and visualization tools
- Retention: 90-day storage, then archive

Data Classification:

Data Type	Sensitivity	Retention	Encryption
Security Alerts	High	"90 days"	"At rest + in transit"
System Logs	Medium	"90 days"	"At rest + in transit"
Application Logs	Low	"30 days"	"In transit only"
Performance Metrics	Low	"90 days"	"None"

Access Control:

- SIEM Platforms: Authentik SSO with MFA required
- API Access: API tokens with 90-day rotation
- Dashboard Access: Role-based permissions (admin, analyst, viewer)
- Alert Management: Admin-only response actions

Encryption Standards:

- Log Transport: TLS 1.3 for all forwarder connections
- At Rest: AES-256 for stored logs and backups
- Certificates: Step-CA issued, auto-renewed

Detection Coverage:

MITRE ATT&CK Technique	Detection Method	Data Source
Initial Access (T1078)	Failed login correlation	Wazuh; pfSense
Execution (T1059)	PowerShell command monitoring	Sysmon

Persistence (T1547)	Registry/startup monitoring	Wazuh FIM
Privilege Escalation (T1068)	Process creation anomalies	Sysmon; Wazuh
Defense Evasion (T1070)	Log clearing detection	Windows Event Log
Credential Access (T1003)	LSASS access monitoring	Sysmon Event ID 10
Discovery (T1082)	System info enumeration	Wazuh; Sysmon
Lateral Movement (T1021)	RDP/SMB connection tracking	Suricata; Wazuh
Collection (T1005)	Unusual file access patterns	Wazuh FIM
Exfiltration (T1041)	Large outbound transfers	pfSense; Suricata
Command and Control (T1071)	Beacon detection	Suricata; pfSense

Use Cases and Scenarios

Practical Use Cases and Detection Scenarios

Scenario 1: Brute Force Attack Detection and Response

Objective: Detect and automatically block SSH brute force attempts

Workflow:

1. Attacker attempts SSH login to docker-vm-01 from 192.168.1.99
2. Failed attempts logged to /var/log/auth.log
3. Splunk Universal Forwarder forwards logs to Splunk indexer
4. Splunk correlation search detects >20 failed attempts in 5 minutes
5. Alert triggered: "SSH Brute Force from 192.168.1.99"
6. Discord webhook notification sent to #security channel
7. Wazuh active response triggers firewall-drop.sh script
8. iptables rule added: Block 192.168.1.99 for 10 minutes
9. Security analyst reviews alert in Splunk dashboard
10. Analyst extends block to permanent if deemed malicious

SPL Query:

```
index=auth sourcetype=syslog "Failed password"
| rex field=_raw "Failed password for (?<user>\w+) from"
| rex field=_raw "(?<src_ip>\d+\.\d+\.\d+\.\d+)"
| stats count by src_ip, dest_host
| where count > 20
| eval threat_level="High", action="Block IP"
```

Scenario 2: Malware Execution Detection

Objective: Detect execution of malicious file on Windows endpoint

Workflow:

1. User on Win11Pro opens malicious email attachment
2. Malware (ransomware.exe) executes
3. Sysmon Event ID 1 (Process Creation) logged
4. Wazuh agent forwards event to Wazuh manager
5. Wazuh rule matches suspicious process characteristics:
 - o Unsigned executable
 - o Launched from %TEMP% directory
 - o Network connection to unknown IP
6. Wazuh generates Critical alert (level 15)
7. Alert forwarded to Splunk for correlation
8. Splunk enriches alert with threat intelligence lookup

9. MD5 hash matches known ransomware family
10. Discord notification: "CRITICAL: Ransomware detected on Win11Pro"
11. Wazuh active response kills process
12. File quarantined to secure location
13. Host isolated from network (manual step)
14. Incident response team engages

Wazuh Rule:

```
<rule id="100200" level="15">
  <if_sid>18106</if_sid>
  <field name="win.eventdata.image">\.exe$</field>
  <field
name="win.eventdata.commandLine">\\Temp\\</field>
  <description>Suspicious executable launched from Temp
directory</description>
  <mitre>
    <id>T1204.002</id>
  </mitre>
</rule>
```

Scenario 3: Lateral Movement Detection

Objective: Identify attacker moving between hosts after initial compromise

Workflow:

1. Attacker compromises Win11Pro via phishing
2. Attacker attempts RDP to DC01 using stolen credentials
3. Suricata IDS detects RDP connection: 192.168.1.200 → 192.168.1.152:3389
4. Windows Security Event 4624 (Successful Logon) on DC01
5. Wazuh agent on DC01 forwards event to manager
6. Splunk correlation search identifies:
 - o User "jdoe" logged in from Win11Pro (unusual source)
 - o Time: 2:30 AM (outside normal business hours)
 - o Logon Type 10 (RemoteInteractive/RDP)
7. Alert generated: "Suspicious Lateral Movement Detected"
8. Discord notification with full context
9. Analyst reviews:
 - o User's normal login pattern (9 AM - 5 PM from office PC)
 - o No scheduled maintenance windows
 - o Source host shows signs of compromise
10. Analyst disables user account in AD
11. Terminates RDP session on DC01
12. Initiates forensic investigation

Splunk Correlation Query:

```
(index=wazuh-alerts rule.id="60204") OR (index=suricata-lan dest_port=3389)
| eval time_hour=strftime(_time,"%H")
| where (time_hour < 6 OR time_hour > 22)
| stats values(src_ip) as source_ips, values(dest_ip) as dest_hosts by user
| where mvcount(dest_hosts) > 1
| eval threat="Lateral movement: user accessed multiple hosts outside business hours"
```

Scenario 4: Configuration Drift Detection

Objective: Detect unauthorized changes to critical system configurations

Workflow:

1. Attacker gains access to web server (Apache LXC)
2. Modifies /etc/ssh/sshd_config to enable PasswordAuthentication
3. Wazuh FIM detects file change within seconds
4. Wazuh generates alert with file diff:
 - o Old value: PasswordAuthentication no
 - o New value: PasswordAuthentication yes
5. Alert includes MD5/SHA256 hash of modified file
6. Alert forwarded to Splunk and displayed in dashboard
7. Discord notification: "File Integrity Violation on Apache LXC"
8. Analyst reviews change:
 - o Change made by root user
 - o No change ticket associated with modification
 - o Unauthorized change confirmed
9. Wazuh active response:
 - o Restores original sshd_config from backup
 - o Restarts SSH service with secure config
10. Analyst investigates how attacker gained root access

Wazuh FIM Alert:

Event: Modified

File: /etc/ssh/sshd_config

Old MD5: a1b2c3d4e5f6...

New MD5: f6e5d4c3b2a1...

User: root

Time: 2024-11-06 14:23:45

Severity: High (level 10)

Scenario 5: Compliance Violation Detection

Objective: Identify hosts failing CIS benchmark requirements

Workflow:

1. Wazuh SCA scan runs every 12 hours
2. Scan detects violations on DC01:
 - o Password minimum length: 8 (Required: 14)
 - o Account lockout threshold: 5 (Required: 3)
 - o Audit log retention: 30 days (Required: 90 days)
3. Wazuh generates compliance report
4. Report forwarded to Splunk
5. Compliance dashboard shows:
 - o Overall score: 89.2% (below 95% target)
 - o Critical failures: 3
 - o Affected hosts: DC01
6. Weekly compliance report emailed to IT management
7. Security team creates remediation tickets:
 - o GPO update for password policy
 - o GPO update for lockout threshold
 - o Increase audit log retention

8. Changes applied via Group Policy
9. Next scan confirms compliance: 96.5%

Wazuh Compliance Query:
Policy: CIS Windows Server 2022
Failed Checks: 3
Score: 89.2%
Risk Level: Medium